DIY_simuling

Design Description

bahaa

DIY_simuling: Design Description

by bahaa

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Chapter 1. Model Version

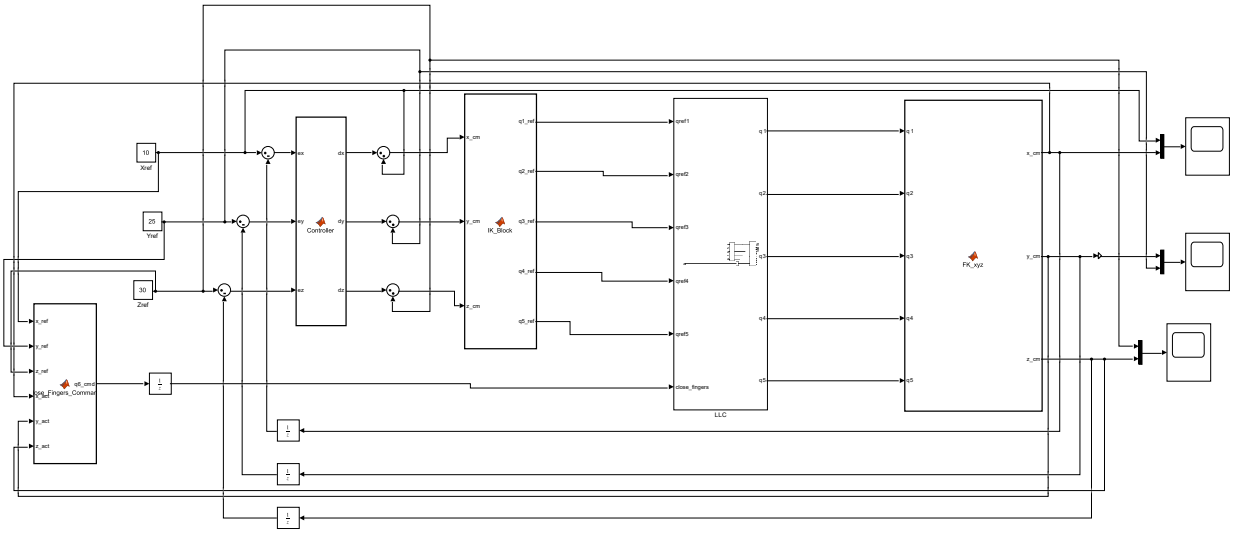
Version: 1.61

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Checksum: 512745293 254321667 3334794620 895490292

Chapter 2. Root System

Figure 2.1. DIY_simuling



Blocks

Parameters

"Gain" (Gain)

Table 2.1. "Gain" Parameters

Parameter	Value
Gain	-1
Multiplication	Element-wise(K.*u)
Parameter minimum	[]
Parameter maximum	[]
Parameter data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule

Parameter	Value
Lock output data type setting against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"MATLAB Function" (MATLAB Function)

Table 2.2. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 2.3. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
x_cm	Input	1	double	1
q1_ref	Output	1	double	1
y_cm	Input	2	double	1
z_cm	Input	3	double	1
q2_ref	Output	2	double	1
q3_ref	Output	3	double	1
q4_ref	Output	4	double	1
q5_ref	Output	5	double	1

MATLAB Function Function Script

```
function [q1_ref, q2_ref, q3_ref, q4_ref, q5_ref] = IK_Block(x_cm, y_cm, z_cm)
```

```
% ============================================================
% IK Block for Simulink
%
% Inputs:
% x_cm, y_cm, z_cm : desired end-effector position [cm]
%
% Outputs:
% q1_ref ... q5_ref : joint reference angles [rad]
%
% Notes:
% - Uses DH-based Generalized IK from base workspace.
% - IK solves DH total angles.
% - theta_offset_DH is subtracted to get real joint angles.
% ============================================================

coder.extrinsic('assignin');
coder.extrinsic('evalin');

% ----------------------------
% Initialize scalar outputs
% ----------------------------
q1_ref = 0;
q2_ref = 0;
q3_ref = 0;
q4_ref = 0;
q5_ref = 0;

% ----------------------------
% Send desired position to base workspace
% ----------------------------
assignin('base', 'ik_x_cm', x_cm);
assignin('base', 'ik_y_cm',-1*y_cm);
assignin('base', 'ik_z_cm', z_cm);

% ----------------------------
% Run IK in base workspace
% ----------------------------
evalin('base', [ ...
    'posTarget_DH.TargetPosition = [ik_x_cm ik_y_cm ik_z_cm] / 100;' ...
    '[qSol_IK, solInfo_IK] = gik_DH(q_prev_IK, posTarget_DH);' ...
    'q_prev_IK = qSol_IK;' ...
    'q_out_IK = qSol_IK - theta_offset_DH;' ...
]);

% ----------------------------
% Read IK result
% q_out_IK is in rad
% ----------------------------
q_tmp = evalin('base', 'q_out_IK');

% ----------------------------
% Separate outputs
% ----------------------------
q1_ref = q_tmp(1);
```

```
q2_ref = q_tmp(2);
q3_ref = q_tmp(3);
q4_ref = q_tmp(4);
q5_ref = q_tmp(5);

end
```

"MATLAB Function1" (MATLAB Function)

Table 2.4. MATLAB Function1 Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 2.5. MATLAB Function1 Argument Summary

Name	Scope	Port	Data Type	Size
q1	Input	1	double	1
x_cm	Output	1	double	1
q2	Input	2	double	1
q3	Input	3	double	1
q4	Input	4	double	1
q5	Input	5	double	1
y_cm	Output	2	double	1
z_cm	Output	3	double	1

MATLAB Function1 Function Script

```
function [x_cm, y_cm, z_cm] = FK_xyz(q1, q2, q3, q4, q5)

% ========================================================
% Forward Kinematics for 5-DOF robotic arm
```

```
%
% Inputs:
% q1..q5 : joint angles [rad]
%
% Outputs:
% x_cm, y_cm, z_cm : end-effector position [cm]
% ============================================================

q = [
    q1;
    q2;
    q3;
    q4;
    q5
];

% ----------------------------
% Robot dimensions [m]
% ----------------------------
a1 = 0.017;
l1 = 0.05;
l2 = 0.12;
l3 = 0.044;
l4 = 0.15;
l5 = 0.152;

% ----------------------------
% Forward kinematics
% Standard DH:
% A = RotZ(theta)*TransZ(d)*TransX(a)*RotX(alpha)
% ----------------------------
T = eye(4);

T = T * DH(a1,  pi/2, l1, q(1));
T = T * DH(l2,  0,    0,  q(2) + pi/2);
T = T * DH(l3,  pi/2, 0,  q(3));
T = T * DH(0,   pi/2, l4, q(4) + pi);
T = T * DH(l5,  0,    0,  q(5) + pi/2);

% Position in meters
p = T(1:3,4);

% Convert to cm
x_cm = p(1) * 100;
y_cm = p(2) * 100;
z_cm = p(3) * 100;

end


% ============================================================
% DH transformation
% ============================================================
function A = DH(a, alpha, d, theta)
```

```
ct = cos(theta);
st = sin(theta);
ca = cos(alpha);
sa = sin(alpha);

A = [
    ct, -st*ca,  st*sa, a*ct;
    st,  ct*ca, -ct*sa, a*st;
    0,      sa,     ca,    d;
    0,       0,      0,    1
];

end
```

Table 2.6. MATLAB Function1 Supporting Functions

Function	Defined By	Path
cos	MATLAB	
eye	MATLAB	
floor	MATLAB	
full	MATLAB	
intmax	MATLAB	
intmin	MATLAB	
isfi	MATLAB	
isfimath	MATLAB	
isinf	MATLAB	
isnumerictype	MATLAB	
issparse	MATLAB	
sin	MATLAB	

"MATLAB Function2" (MATLAB Function)

Table 2.7. MATLAB Function2 Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1

Property	Value
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 2.8. MATLAB Function2 Argument Summary

Name	Scope	Port	Data Type	Size
ex	Input	1	double	1
dx	Output	1	double	1
ey	Input	2	double	1
ez	Input	3	double	1
dy	Output	2	double	1
dz	Output	3	double	1

MATLAB Function2 Function Script

```
function [dx, dy, dz] = Controller(ex, ey, ez)

dx = 0;
dy = 0;
dz = 0;

end
```

"MATLAB Function3" (MATLAB Function)

Table 2.9. MATLAB Function3 Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point

Property	Value
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 2.10. MATLAB Function3 Argument Summary

Name	Scope	Port	Data Type	Size
x_ref	Input	1	double	1
q6_cmd	Output	1	double	1
y_ref	Input	2	double	1
z_ref	Input	3	double	1
x_act	Input	4	double	1
y_act	Input	5	double	1
z_act	Input	6	double	1

MATLAB Function3 Function Script

```
function q6_cmd = Close_Fingers_Command(x_ref, y_ref, z_ref, x_act, y_act,
 z_act)

% Error in cm
ex = x_ref - x_act;
ey = y_ref - y_act;
ez = z_ref - z_act;

e_norm = sqrt(ex^2 + ey^2 + ez^2);

% reaching tolerance [cm]
tol_cm = 0.5;

if e_norm < tol_cm
    q6_cmd = 0.022;   % close
else
    q6_cmd = 0;       % open
end

end
```

Table 2.11. MATLAB Function3 Supporting Functions

Function	Defined By	Path
full	MATLAB	
intmax	MATLAB	
intmin	MATLAB	

Function	Defined By	Path
isfi	MATLAB	
isfimath	MATLAB	
ismatrix	MATLAB	
isnan	MATLAB	
isnumerictype	MATLAB	
issparse	MATLAB	
mpower	MATLAB	
power	MATLAB	
sqrt	MATLAB	

"Mux" (Mux)

Table 2.12. "Mux" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

"Mux1" (Mux)

Table 2.13. "Mux1" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

"Mux2" (Mux)

Table 2.14. "Mux2" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

"Sum" (Sum)

Table 2.15. "Sum" Parameters

Parameter	Value
Icon shape	round
List of signs	|+-
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Sum1" (Sum)

Table 2.16. "Sum1" Parameters

Parameter	Value
Icon shape	round
List of signs	|+-
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Sum2" (Sum)

Table 2.17. "Sum2" Parameters

Parameter	Value
Icon shape	round
List of signs	|+-
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Sum3" (Sum)

Table 2.18. "Sum3" Parameters

Parameter	Value
Icon shape	round
List of signs	|++
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Sum4" (Sum)

Table 2.19. "Sum4" Parameters

Parameter	Value
Icon shape	round
List of signs	|++
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Sum5" (Sum)

Table 2.20. "Sum5" Parameters

Parameter	Value
Icon shape	round
List of signs	|++
Sum over	All dimensions
Dimension	1
Require all inputs to have the same data type	off
Accumulator data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Unit Delay" (UnitDelay)

Table 2.21. "Unit Delay" Parameters

Parameter	Value
Initial condition	0
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Unit Delay1" (UnitDelay)

Table 2.22. "Unit Delay1" Parameters

Parameter	Value
Initial condition	0
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Unit Delay2" (UnitDelay)

Table 2.23. "Unit Delay2" Parameters

Parameter	Value
Initial condition	0
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Unit Delay3" (UnitDelay)

Table 2.24. "Unit Delay3" Parameters

Parameter	Value
Initial condition	0
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1

Parameter	Value
State name must resolve to Simulink signal object	off

"Xref" (Constant)

Table 2.25. "Xref" Parameters

Parameter	Value
Constant value	10
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit from 'Constant value'
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

"Yref" (Constant)

Table 2.26. "Yref" Parameters

Parameter	Value
Constant value	25
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit from 'Constant value'
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

"Zref" (Constant)

Table 2.27. "Zref" Parameters

Parameter	Value
Constant value	30
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit from 'Constant value'
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

Block Execution Order

1. Xref (Constant)
2. Unit Delay3 (UnitDelay)
3. Integrator (Integrator)
4. *MATLAB Function*
    1. SFunction (S-Function)
5. K6 (Gain)
6. INPUT_1_1_1 (SimscapeInputBlock)
7. Unit Delay1 (UnitDelay)
8. Sum (Sum)
9. Yref (Constant)
10. Unit Delay2 (UnitDelay)
11. Sum1 (Sum)
12. Zref (Constant)
13. Unit Delay (UnitDelay)
14. Sum2 (Sum)
15. *MATLAB Function2*
    1. SFunction (S-Function)
16. Sum3 (Sum)
17. Sum4 (Sum)
18. Sum5 (Sum)
19. *MATLAB Function*
    1. SFunction (S-Function)
20. Integrator (Integrator)
21. *MATLAB Function*
    1. TmpSignal ConversionAt SFunction Inport1 (SignalConversion)
    2. SFunction (S-Function)

22. INPUT_3_1_1 (SimscapeInputBlock)
23. INPUT_2_1_1 (SimscapeInputBlock)
24. INPUT_4_1_1 (SimscapeInputBlock)
25. INPUT_5_1_1 (SimscapeInputBlock)
26. INPUT_6_1_1 (SimscapeInputBlock)
27. STATE_1 (SimscapeExecutionBlock)
28. OUTPUT_1_0 (SimscapeExecutionBlock)
29. *MATLAB Function1*
    1. SFunction (S-Function)
30. Scope (Scope)
31. Gain (Gain)
32. Scope1 (Scope)
33. Scope2 (Scope)
34. input (PMIOPort)
35. input (PMIOPort)
36. input (PMIOPort)
37. input (PMIOPort)
38. input (PMIOPort)
39. input (PMIOPort)
40. input (PMIOPort)
41. output (PMIOPort)
42. ReferenceFrame (SimscapeMultibodyBlock)
43. Solid (SimscapeMultibodyBlock)
44. Transform (SimscapeMultibodyBlock)
45. Transform1 (SimscapeMultibodyBlock)
46. F1 (PMIOPort)
47. F (PMIOPort)
48. ReferenceFrame (SimscapeMultibodyBlock)
49. Solid (SimscapeMultibodyBlock)
50. Transform (SimscapeMultibodyBlock)
51. Transform1 (SimscapeMultibodyBlock)
52. F1 (PMIOPort)
53. F (PMIOPort)
54. ReferenceFrame (SimscapeMultibodyBlock)
55. Solid (SimscapeMultibodyBlock)
56. Transform (SimscapeMultibodyBlock)
57. Transform1 (SimscapeMultibodyBlock)
58. F1 (PMIOPort)
59. F (PMIOPort)
60. ReferenceFrame (SimscapeMultibodyBlock)
61. Solid (SimscapeMultibodyBlock)
62. Transform (SimscapeMultibodyBlock)
63. Transform1 (SimscapeMultibodyBlock)
64. F1 (PMIOPort)
65. F (PMIOPort)
66. ReferenceFrame (SimscapeMultibodyBlock)
67. Solid (SimscapeMultibodyBlock)
68. Transform (SimscapeMultibodyBlock)
69. Transform1 (SimscapeMultibodyBlock)

70. F1 (PMIOPort)
71. F (PMIOPort)
72. ReferenceFrame (SimscapeMultibodyBlock)
73. Solid (SimscapeMultibodyBlock)
74. Transform (SimscapeMultibodyBlock)
75. Transform1 (SimscapeMultibodyBlock)
76. F1 (PMIOPort)
77. F (PMIOPort)
78. ReferenceFrame (SimscapeMultibodyBlock)
79. Solid (SimscapeMultibodyBlock)
80. Transform (SimscapeMultibodyBlock)
81. Transform1 (SimscapeMultibodyBlock)
82. Transform2 (SimscapeMultibodyBlock)
83. Transform3 (SimscapeMultibodyBlock)
84. F (PMIOPort)
85. F3 (PMIOPort)
86. F2 (PMIOPort)
87. F1 (PMIOPort)
88. ReferenceFrame (SimscapeMultibodyBlock)
89. Solid (SimscapeMultibodyBlock)
90. Transform (SimscapeMultibodyBlock)
91. Transform1 (SimscapeMultibodyBlock)
92. Transform2 (SimscapeMultibodyBlock)
93. Transform3 (SimscapeMultibodyBlock)
94. F (PMIOPort)
95. F3 (PMIOPort)
96. F2 (PMIOPort)
97. F1 (PMIOPort)
98. ReferenceFrame (SimscapeMultibodyBlock)
99. Solid (SimscapeMultibodyBlock)
100. Transform (SimscapeMultibodyBlock)
101. Transform1 (SimscapeMultibodyBlock)
102. Transform2 (SimscapeMultibodyBlock)
103. Transform3 (SimscapeMultibodyBlock)
104. F3 (PMIOPort)
105. F2 (PMIOPort)
106. F1 (PMIOPort)
107. F (PMIOPort)
108. ReferenceFrame (SimscapeMultibodyBlock)
109. Solid (SimscapeMultibodyBlock)
110. Transform (SimscapeMultibodyBlock)
111. F (PMIOPort)
112. Cylindrical Joint (SimscapeMultibodyBlock)
113. Revolute Joint (SimscapeMultibodyBlock)
114. Revolute Joint1 (SimscapeMultibodyBlock)
115. Revolute Joint2 (SimscapeMultibodyBlock)
116. Revolute1 (SimscapeMultibodyBlock)
117. Revolute10 (SimscapeMultibodyBlock)
118. Revolute11 (SimscapeMultibodyBlock)

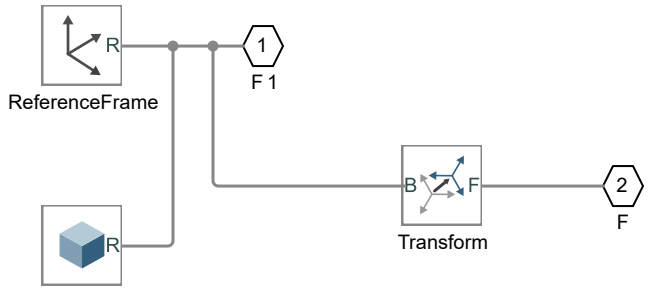
119. Revolute6 (SimscapeMultibodyBlock)
120. Revolute7 (SimscapeMultibodyBlock)
121. Revolute8 (SimscapeMultibodyBlock)
122. Revolute9 (SimscapeMultibodyBlock)
123. Weld Joint (SimscapeMultibodyBlock)
124. Conn1 (PMIOPort)
125. input (PMIOPort)
126. input (PMIOPort)
127. input (PMIOPort)
128. input (PMIOPort)
129. input (PMIOPort)
130. output (PMIOPort)
131. output (PMIOPort)
132. output (PMIOPort)
133. output (PMIOPort)
134. output (PMIOPort)
135. ReferenceFrame (SimscapeMultibodyBlock)
136. Solid (SimscapeMultibodyBlock)
137. Transform (SimscapeMultibodyBlock)
138. F1 (PMIOPort)
139. F (PMIOPort)
140. ReferenceFrame (SimscapeMultibodyBlock)
141. Solid (SimscapeMultibodyBlock)
142. Transform (SimscapeMultibodyBlock)
143. Transform1 (SimscapeMultibodyBlock)
144. F (PMIOPort)
145. F1 (PMIOPort)
146. ReferenceFrame (SimscapeMultibodyBlock)
147. Solid (SimscapeMultibodyBlock)
148. Transform (SimscapeMultibodyBlock)
149. Transform1 (SimscapeMultibodyBlock)
150. Transform2 (SimscapeMultibodyBlock)
151. F1 (PMIOPort)
152. F (PMIOPort)
153. F2 (PMIOPort)
154. ReferenceFrame (SimscapeMultibodyBlock)
155. Solid (SimscapeMultibodyBlock)
156. Transform (SimscapeMultibodyBlock)
157. Transform1 (SimscapeMultibodyBlock)
158. F1 (PMIOPort)
159. F (PMIOPort)
160. ReferenceFrame (SimscapeMultibodyBlock)
161. Solid (SimscapeMultibodyBlock)
162. Transform (SimscapeMultibodyBlock)
163. F (PMIOPort)
164. ReferenceFrame (SimscapeMultibodyBlock)
165. Solid (SimscapeMultibodyBlock)
166. Transform (SimscapeMultibodyBlock)
167. Transform1 (SimscapeMultibodyBlock)

168. F (PMIOPort)
169. F1 (PMIOPort)
170. SINK_1 (SimscapeSinkBlock)
171. a (PMIOPort)
172. ReferenceFrame (SimscapeMultibodyBlock)
173. Solid (SimscapeMultibodyBlock)
174. Transform (SimscapeMultibodyBlock)
175. Transform1 (SimscapeMultibodyBlock)
176. F1 (PMIOPort)
177. F (PMIOPort)
178. J1 (SimscapeMultibodyBlock)
179. J2 (SimscapeMultibodyBlock)
180. J3 (SimscapeMultibodyBlock)
181. J4 (SimscapeMultibodyBlock)
182. J5 (SimscapeMultibodyBlock)
183. MechanismConfiguration (SimscapeMultibodyBlock)
184. Transform (SimscapeMultibodyBlock)
185. World (SimscapeMultibodyBlock)
186. *MATLAB Function3*
    1. SFunction (S-Function)

Chapter 3. Subsystems

Base

Figure 3.1. DIY_simuling/LLC/Robot_Plant/Base



Blocks

Parameters

"F" (PMIOPort)

Table 3.1. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.2. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.3. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.4. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(1).mass
	kg
	compiletime
	smiData.Solid(1).CoM
	mm
	compiletime
	smiData.Solid(1).MoI
	kg*mm^2
	compiletime
	smiData.Solid(1).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(1).color

Parameter	Value
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(1).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m

Parameter	Value
	compiletime
	STEP
	Robot - Base.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

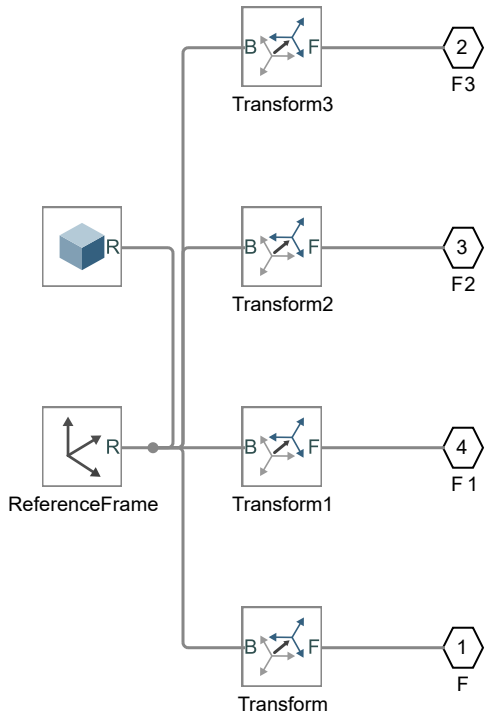
Table 3.5. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(9).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(9).angle

Parameter	Value
	compiletime
	smiData.RigidTransform(9).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

cap

Figure 3.2. DIY_simuling/LLC/Robot_Plant/EE/cap



Blocks

Parameters

"F" (PMIOPort)

Table 3.6. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.7. "F1" Parameters

Parameter	Value
Port number	4
Port location on parent subsystem	Right

"F2" (PMIOPort)

Table 3.8. "F2" Parameters

Parameter	Value
Port number	3
Port location on parent subsystem	Right

"F3" (PMIOPort)

Table 3.9. "F3" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.10. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.11. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(7).mass
	kg

Parameter	Value
	compiletime
	smiData.Solid(7).CoM
	mm
	compiletime
	smiData.Solid(7).MoI
	kg*mm^2
	compiletime
	smiData.Solid(7).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(7).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(7).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime

Parameter	Value
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J8 cap.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.12. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(6).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(6).angle
	compiletime
	smiData.RigidTransform(6).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

Table 3.13. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(27).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(27).angle
	compiletime
	smiData.RigidTransform(27).axis
	compiletime
	+X
	+Y
	+Y
	+Z

Parameter	Value
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform2" (SimscapeMultibodyBlock)

Table 3.14. "Transform2" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(29).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(29).angle
	compiletime
	smiData.RigidTransform(29).axis

Parameter	Value
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform3" (SimscapeMultibodyBlock)

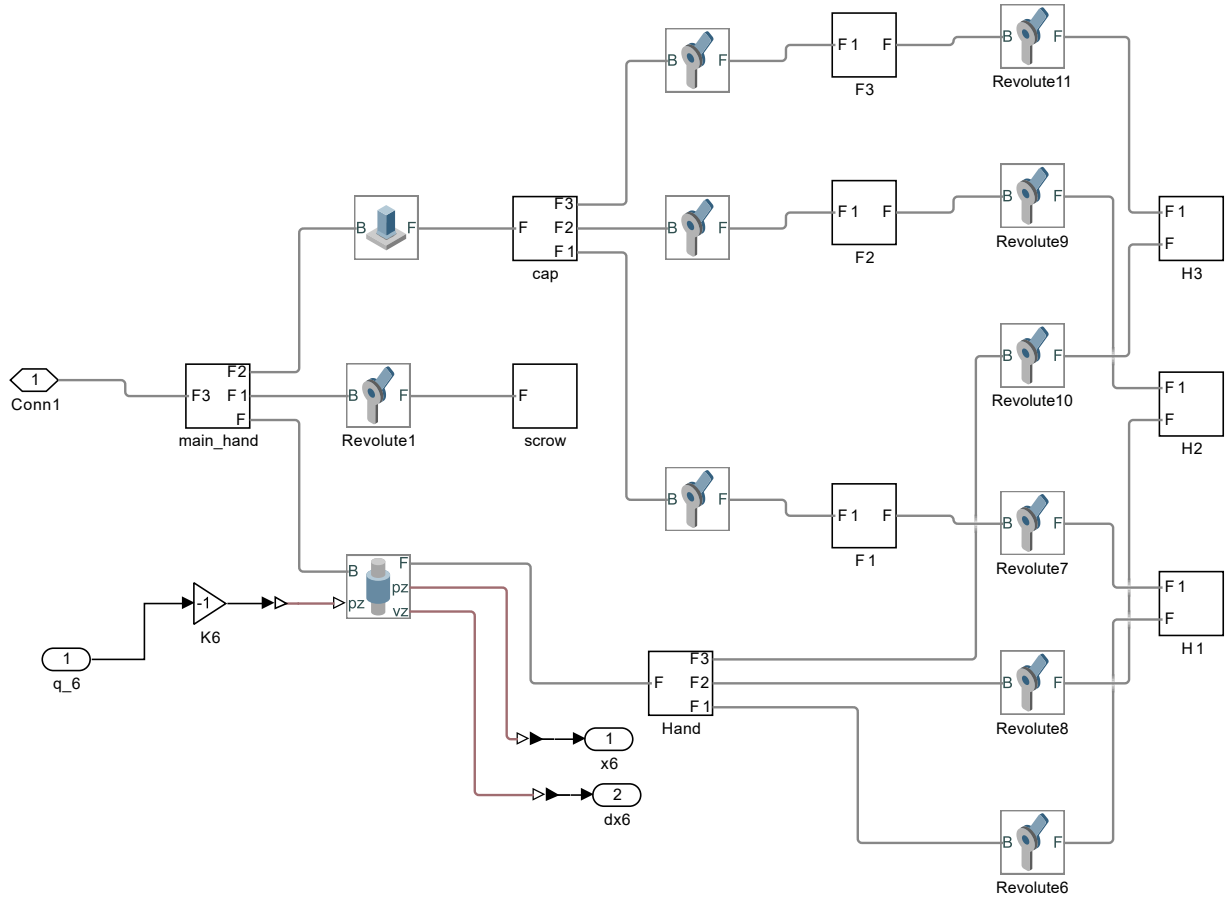
Table 3.15. "Transform3" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(31).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis

Parameter	Value
	rad
	+Z
	smiData.RigidTransform(31).angle
	compiletime
	smiData.RigidTransform(31).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

EE

Figure 3.3. DIY_simuling/LLC/Robot_Plant/EE



Blocks

Parameters

"Conn1" (PMIOPort)

Table 3.16. "Conn1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"Cylindrical Joint" (SimscapeMultibodyBlock)

Table 3.17. "Cylindrical Joint" Parameters

Parameter	Value
	CylindricalJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	m
	compiletime
	0
	N/m
	compiletime
	0
	N/(m/s)
	compiletime
	ComputedTorque
	InputMotion
	on
	on
	off

Parameter	Value
	off
	off
	off
	on
	0
	m
	compiletime
	1e6
	N/m
	compiletime
	1e3
	N/(m/s)
	compiletime
	1e-4
	m
	compiletime
	on
	0.022
	m
	compiletime
	1e6
	N/m
	compiletime
	1e3
	N/(m/s)
	compiletime
	1e-4
	m
	compiletime
	off
	High
	0
	m
	compiletime
	off
	High
	0

Parameter	Value
	m/s
	compiletime
	0
	deg
	compiletime
	0
	N*m/deg
	compiletime
	0
	N*m/(deg/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	on
	-1
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	1
	deg
	compiletime
	1e4

Parameter	Value
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	High
	0
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.cylindrical_joint

"dx6" (Outport)

Table 3.18. "dx6" Parameters

Parameter	Value
Port number	2
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off

Parameter	Value
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"K6" (Gain)

Table 3.19. "K6" Parameters

Parameter	Value
Gain	-1
Multiplication	Element-wise(K.*u)
Parameter minimum	[]
Parameter maximum	[]
Parameter data type	Inherit: Inherit via internal rule
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via internal rule
Lock output data type setting against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"PS-Simulink Converter1" (SubSystem)

Table 3.20. "PS-Simulink Converter1" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off

Parameter	Value
Frequencies	[]
Unit	m/s
AffineConversion	off

"PS-Simulink Converter5" (SubSystem)

Table 3.21. "PS-Simulink Converter5" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	m
AffineConversion	off

"q_6" (Inport)

Table 3.22. "q_6" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"Revolute Joint" (SimscapeMultibodyBlock)

Table 3.23. "Revolute Joint" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	deg
	compiletime
	0
	N*m/deg
	compiletime
	0
	N*m/(deg/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4

Parameter	Value
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	High
	0
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute Joint1" (SimscapeMultibodyBlock)

Table 3.24. "Revolute Joint1" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	deg
	compiletime
	0
	N*m/deg
	compiletime
	0
	N*m/(deg/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	High
	0
	deg
	compiletime
	off
	High
	0

Parameter	Value
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute Joint2" (SimscapeMultibodyBlock)

Table 3.25. "Revolute Joint2" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	deg
	compiletime
	0
	N*m/deg
	compiletime
	0
	N*m/(deg/s)
	compiletime

Parameter	Value
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	High
	0

Parameter	Value
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute1" (SimscapeMultibodyBlock)

Table 3.26. "Revolute1" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0

Parameter	Value
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1

Parameter	Value
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(2).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute10" (SimscapeMultibodyBlock)

Table 3.27. "Revolute10" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase

Parameter	Value
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg

Parameter	Value
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(11).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute11" (SimscapeMultibodyBlock)

Table 3.28. "Revolute11" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off

Parameter	Value
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(12).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute6" (SimscapeMultibodyBlock)

Table 3.29. "Revolute6" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)

Parameter	Value
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(7).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute7" (SimscapeMultibodyBlock)

Table 3.30. "Revolute7" Parameters

Parameter	Value
	RevoluteJoint
	off
	off

Parameter	Value
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime

Parameter	Value
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(8).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute8" (SimscapeMultibodyBlock)

Table 3.31. "Revolute8" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	NoTorque
	ComputedMotion
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(9).Rz.Pos
	deg
	compiletime
	off
	High
	0

Parameter	Value
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Revolute9" (SimscapeMultibodyBlock)

Table 3.32. "Revolute9" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime

Parameter	Value
	NoTorque
	ComputedMotion
	off
	off
	off
	off
	off
	off
	off
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	off
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(10).Rz.Pos

Parameter	Value
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"Simulink-PS Converter6" (SubSystem)

Table 3.33. "Simulink-PS Converter6" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	m
AffineConversion	off
FilteringAndDerivatives	filter
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001
UdotUserProvided	0

"Weld Joint" (SimscapeMultibodyBlock)

Table 3.34. "Weld Joint" Parameters

Parameter	Value
	WeldJoint
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	simmechanics.library.joints.weld_joint

"x6" (Outport)

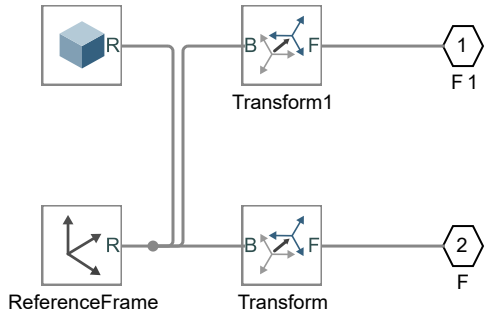
Table 3.35. "x6" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]

Parameter	Value
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

F1

Figure 3.4. DIY_simuling/LLC/Robot_Plant/EE/F1



Blocks

Parameters

"F" (PMIOPort)

Table 3.36. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.37. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.38. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.39. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(12).mass
	kg
	compiletime
	smiData.Solid(12).CoM
	mm
	compiletime
	smiData.Solid(12).MoI
	kg*mm^2
	compiletime
	smiData.Solid(12).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(12).color

Parameter	Value
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(12).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m

Parameter	Value
	compiletime
	STEP
	Robot - J10 finger.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.40. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(17).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(17).angle

Parameter	Value
	compiletime
	smiData.RigidTransform(17).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

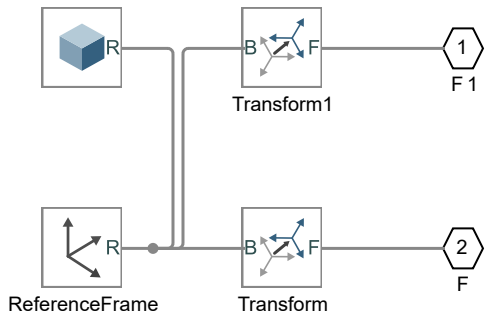
Table 3.41. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(28).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg

Parameter	Value
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(28).angle
	compiletime
	smiData.RigidTransform(28).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

F2

Figure 3.5. DIY_simuling/LLC/Robot_Plant/EE/F2



Blocks

Parameters

"F" (PMIOPort)

Table 3.42. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.43. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.44. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.45. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(12).mass
	kg
	compiletime
	smiData.Solid(12).CoM
	mm

Parameter	Value
	compiletime
	smiData.Solid(12).MoI
	kg*mm^2
	compiletime
	smiData.Solid(12).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(12).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(12).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime

Parameter	Value
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J10 finger.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.46. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian

Parameter	Value
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(21).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(21).angle
	compiletime
	smiData.RigidTransform(21).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

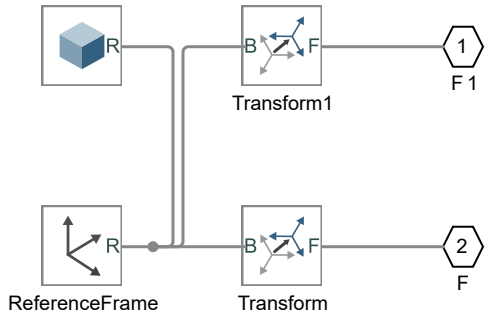
Table 3.47. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(30).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(30).angle
	compiletime
	smiData.RigidTransform(30).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

F3

Figure 3.6. DIY_simuling/LLC/Robot_Plant/EE/F3



Blocks

Parameters

"F" (PMIOPort)

Table 3.48. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.49. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.50. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.51. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(12).mass
	kg
	compiletime
	smiData.Solid(12).CoM
	mm
	compiletime
	smiData.Solid(12).MoI
	kg*mm^2
	compiletime
	smiData.Solid(12).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(12).color

Parameter	Value
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(12).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m

Parameter	Value
	compiletime
	STEP
	Robot - J10 finger.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.52. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(25).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(25).angle

Parameter	Value
	compiletime
	smiData.RigidTransform(25).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

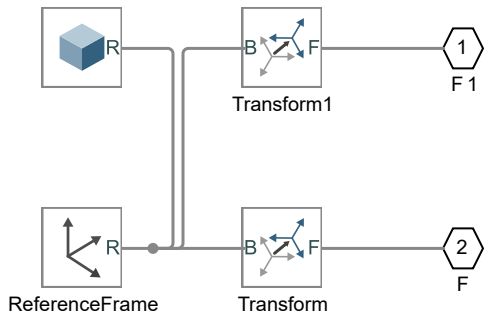
Table 3.53. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(32).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg

Parameter	Value
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(32).angle
	compiletime
	smiData.RigidTransform(32).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

H1

Figure 3.7. DIY_simuling/LLC/Robot_Plant/EE/H1



Blocks

Parameters

"F" (PMIOPort)

Table 3.54. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.55. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.56. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.57. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(8).mass
	kg
	compiletime
	smiData.Solid(8).CoM
	mm

Parameter	Value
	compiletime
	smiData.Solid(8).MoI
	kg*mm^2
	compiletime
	smiData.Solid(8).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(8).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(8).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime

Parameter	Value
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J9 hanger_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.58. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian

Parameter	Value
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(16).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(16).angle
	compiletime
	smiData.RigidTransform(16).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

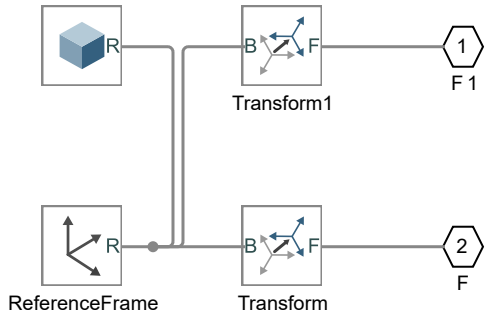
Table 3.59. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(18).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(18).angle
	compiletime
	smiData.RigidTransform(18).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

H2

Figure 3.8. DIY_simuling/LLC/Robot_Plant/EE/H2



Blocks

Parameters

"F" (PMIOPort)

Table 3.60. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.61. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.62. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.63. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(8).mass
	kg
	compiletime
	smiData.Solid(8).CoM
	mm
	compiletime
	smiData.Solid(8).MoI
	kg*mm^2
	compiletime
	smiData.Solid(8).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(8).color

Parameter	Value
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(8).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m

Parameter	Value
	compiletime
	STEP
	Robot - J9 hanger_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.64. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(20).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(20).angle

Parameter	Value
	compiletime
	smiData.RigidTransform(20).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

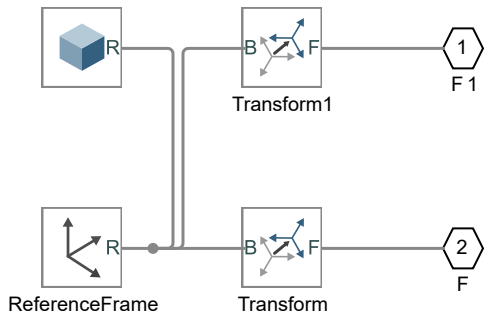
Table 3.65. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(22).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg

Parameter	Value
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(22).angle
	compiletime
	smiData.RigidTransform(22).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

H3

Figure 3.9. DIY_simuling/LLC/Robot_Plant/EE/H3



Blocks

Parameters

"F" (PMIOPort)

Table 3.66. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.67. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.68. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.69. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(8).mass
	kg
	compiletime
	smiData.Solid(8).CoM
	mm

Parameter	Value
	compiletime
	smiData.Solid(8).MoI
	kg*mm^2
	compiletime
	smiData.Solid(8).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(8).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(8).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime

Parameter	Value
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J9 hanger_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.70. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian

Parameter	Value
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(24).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(24).angle
	compiletime
	smiData.RigidTransform(24).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

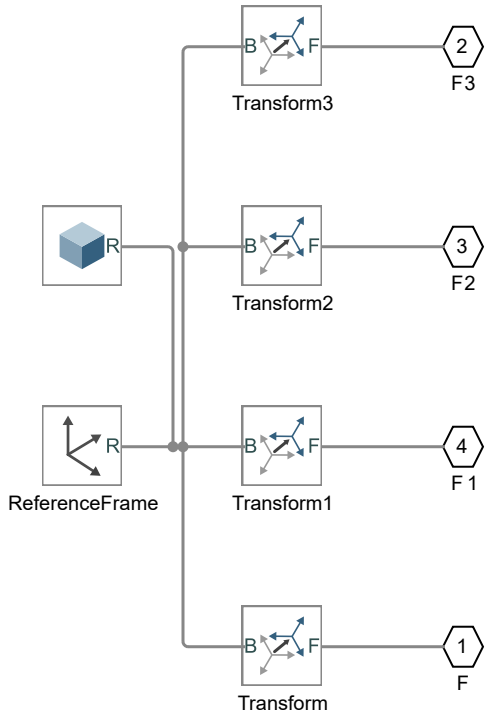
Table 3.71. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(26).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(26).angle
	compiletime
	smiData.RigidTransform(26).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Hand

Figure 3.10. DIY_simuling/LLC/Robot_Plant/EE/Hand



Blocks

Parameters

"F" (PMIOPort)

Table 3.72. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.73. "F1" Parameters

Parameter	Value
Port number	4
Port location on parent subsystem	Right

"F2" (PMIOPort)

Table 3.74. "F2" Parameters

Parameter	Value
Port number	3
Port location on parent subsystem	Right

"F3" (PMIOPort)

Table 3.75. "F3" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.76. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.77. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(11).mass
	kg

Parameter	Value
	compiletime
	smiData.Solid(11).CoM
	mm
	compiletime
	smiData.Solid(11).MoI
	kg*mm^2
	compiletime
	smiData.Solid(11).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(11).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(11).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime

Parameter	Value
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J6 hand.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.78. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(2).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(2).angle
	compiletime
	smiData.RigidTransform(2).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

Table 3.79. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(15).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(15).angle
	compiletime
	smiData.RigidTransform(15).axis
	compiletime
	+X
	+Y
	+Y
	+Z

Parameter	Value
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform2" (SimscapeMultibodyBlock)

Table 3.80. "Transform2" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(19).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(19).angle
	compiletime
	smiData.RigidTransform(19).axis

Parameter	Value
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform3" (SimscapeMultibodyBlock)

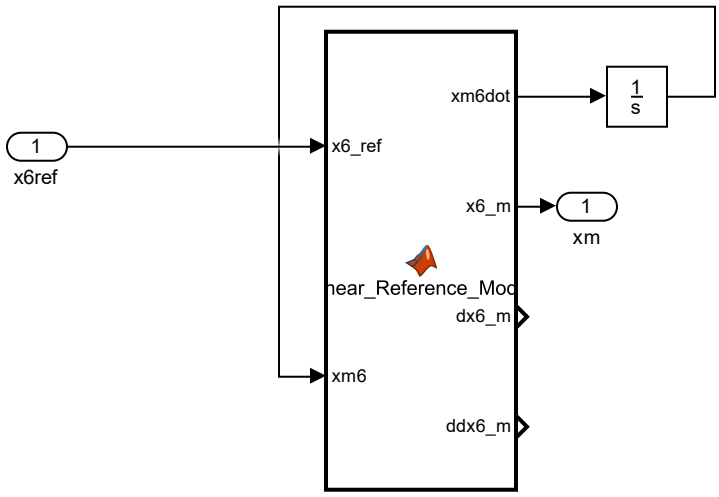
Table 3.81. "Transform3" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(23).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis

Parameter	Value
	rad
	+Z
	smiData.RigidTransform(23).angle
	compiletime
	smiData.RigidTransform(23).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Linear Joint Reference Model

Figure 3.11. DIY_simuling/LLC/Linear Joint Reference Model



Blocks

Parameters

"Integrator" (Integrator)

Table 3.82. "Integrator" Parameters

Parameter	Value
External reset	none
Initial condition source	internal
Initial condition	[0;0]
Limit output	off
Upper saturation limit	inf
Lower saturation limit	-inf
Wrap state	off
Wrapped state upper value	pi
Wrapped state lower value	-pi
Show saturation port	off
Show state port	off
Ignore limit and reset when linearizing	off
Enable zero-crossing detection	on
State Name (e.g., 'position')	''

"MATLAB Function" (MATLAB Function)

Table 3.83. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default

Property	Value
Input fi math	fimath(...)
Description	

Table 3.84. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
x6_ref	Input	1	double	1
xm6dot	Output	1	double	2
xm6	Input	2	double	2
x6_m	Output	2	double	1
dx6_m	Output	3	double	1
ddx6_m	Output	4	double	1

MATLAB Function Function Script

```
function [xm6dot, x6_m, dx6_m, ddx6_m] = Linear_Reference_Model(x6_ref, xm6)

% ============================================================
% Second-order reference model for linear joint x6
% Range: 0 <= x6 <= 0.022 m
% ============================================================

% Force scalar reference
x6_ref = min(max(x6_ref, 0), 0.022);

% State vector
xm6 = reshape(xm6, [2,1]);

% Parameters
zeta = 1;
wn   = 2;

% State-space model
A = [
    0, 1;
   -wn^2, -2*zeta*wn
];

B = [
    0;
    wn^2
];

% Dynamics
xm6dot = A*xm6 + B*x6_ref;

% Outputs
```

```
x6_m   = xm6(1);
dx6_m  = xm6(2);
ddx6_m = xm6dot(2);

end
```

Table 3.85. MATLAB Function Supporting Functions

Function	Defined By	Path
floor	MATLAB	
full	MATLAB	
intmax	MATLAB	
intmin	MATLAB	
isfi	MATLAB	
isfimath	MATLAB	
isinf	MATLAB	
ismatrix	MATLAB	
isnan	MATLAB	
isnumerictype	MATLAB	
issparse	MATLAB	
max	MATLAB	
min	MATLAB	
mpower	MATLAB	
power	MATLAB	

"x6ref" (Inport)

Table 3.86. "x6ref" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

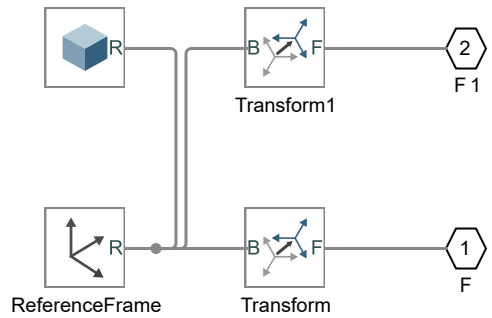
"xm" (Outport)

Table 3.87. "xm" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

Link1

Figure 3.12. DIY_simuling/LLC/Robot_Plant/Link1



Blocks

Parameters

"F" (PMIOPort)

Table 3.88. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.89. "F1" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.90. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.91. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(2).mass
	kg
	compiletime
	smiData.Solid(2).CoM
	mm

Parameter	Value
	compiletime
	smiData.Solid(2).MoI
	kg*mm^2
	compiletime
	smiData.Solid(2).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(2).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(2).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime

Parameter	Value
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J1.stp.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.92. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian

Parameter	Value
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(10).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(10).angle
	compiletime
	smiData.RigidTransform(10).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

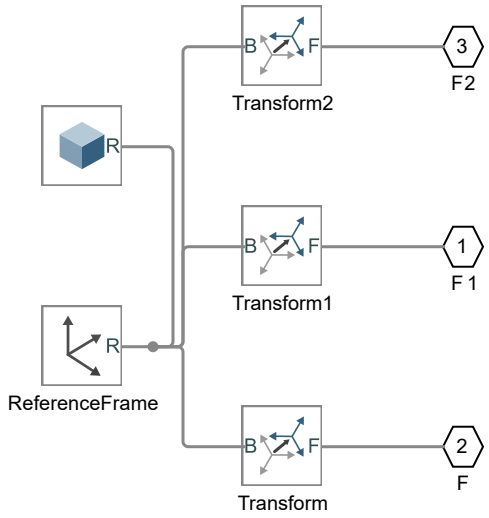
Table 3.93. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(11).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(11).angle
	compiletime
	smiData.RigidTransform(11).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Link1_2

Figure 3.13. DIY_simuling/LLC/Robot_Plant/Link1_2



Blocks

Parameters

"F" (PMIOPort)

Table 3.94. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.95. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"F2" (PMIOPort)

Table 3.96. "F2" Parameters

Parameter	Value
Port number	3
Port location on parent subsystem	Right

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.97. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.98. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(3).mass
	kg
	compiletime
	smiData.Solid(3).CoM
	mm
	compiletime
	smiData.Solid(3).MoI
	kg*mm^2
	compiletime

Parameter	Value
	smiData.Solid(3).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(3).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(3).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1

Parameter	Value
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J2 (LEFT).stp.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.99. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z

Parameter	Value
	smiData.RigidTransform(12).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(12).angle
	compiletime
	smiData.RigidTransform(12).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

Table 3.100. "Transform1" Parameters

Parameter	Value
	RigidTransform

Parameter	Value
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(13).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(13).angle
	compiletime
	smiData.RigidTransform(13).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform2" (SimscapeMultibodyBlock)

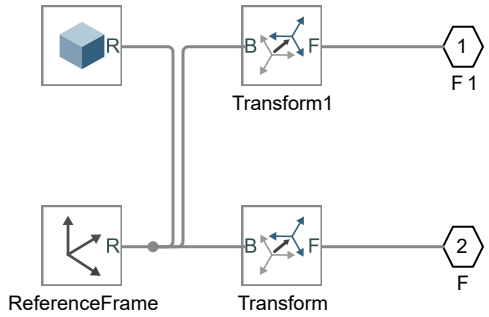
Table 3.101. "Transform2" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(33).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(33).angle
	compiletime
	smiData.RigidTransform(33).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Link1_4

Figure 3.14. DIY_simuling/LLC/Robot_Plant/Link1_4



Blocks

Parameters

"F" (PMIOPort)

Table 3.102. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.103. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.104. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.105. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(6).mass
	kg
	compiletime
	smiData.Solid(6).CoM
	mm
	compiletime
	smiData.Solid(6).MoI
	kg*mm^2
	compiletime
	smiData.Solid(6).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(6).color

Parameter	Value
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(6).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m

Parameter	Value
	compiletime
	STEP
	Robot - J4 (ARM).stp.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.106. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(35).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(35).angle

Parameter	Value
	compiletime
	smiData.RigidTransform(35).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

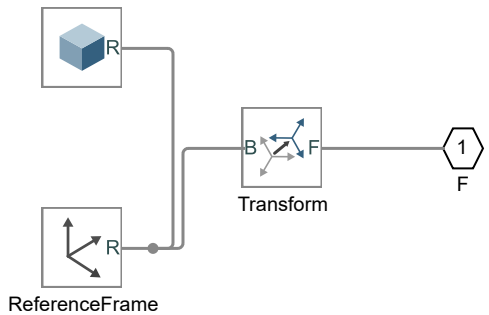
Table 3.107. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(38).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg

Parameter	Value
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(38).angle
	compiletime
	smiData.RigidTransform(38).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Link2_2

Figure 3.15. DIY_simuling/LLC/Robot_Plant/Link2_2



Blocks

Parameters

"F" (PMIOPort)

Table 3.108. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.109. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.110. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(5).mass
	kg
	compiletime
	smiData.Solid(5).CoM
	mm
	compiletime
	smiData.Solid(5).MoI
	kg*mm^2
	compiletime
	smiData.Solid(5).PoI
	kg*mm^2
	compiletime

Parameter	Value
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(5).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(5).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1

Parameter	Value
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J2 (RIGHT).stp.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

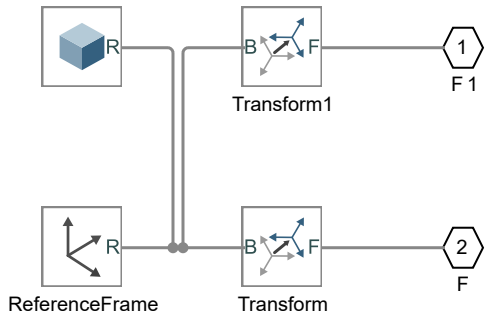
Table 3.111. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(14).translation
	compiletime
	0

Parameter	Value
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(14).angle
	compiletime
	smiData.RigidTransform(14).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

link2_4

Figure 3.16. DIY_simuling/LLC/Robot_Plant/link2_4



Blocks

Parameters

"F" (PMIOPort)

Table 3.112. "F" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.113. "F1" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.114. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame

Parameter	Value
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.115. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(13).mass
	kg
	compiletime
	smiData.Solid(13).CoM
	mm
	compiletime
	smiData.Solid(13).MoI
	kg*mm^2
	compiletime
	smiData.Solid(13).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(13).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]

Parameter	Value
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(13).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J4 (J6 JOINT).stp.step_Default_sldprt.STEP
	m

Parameter	Value
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.116. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(7).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(7).angle
	compiletime
	smiData.RigidTransform(7).axis
	compiletime
	+X

Parameter	Value
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

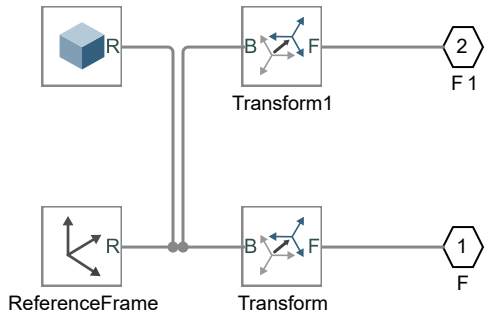
Table 3.117. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(36).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z

Parameter	Value
	smiData.RigidTransform(36).angle
	compiletime
	smiData.RigidTransform(36).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Link3

Figure 3.17. DIY_simuling/LLC/Robot_Plant/Link3



Blocks

Parameters

"F" (PMIOPort)

Table 3.118. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"F1" (PMIOPort)

Table 3.119. "F1" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.120. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.121. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(4).mass
	kg
	compiletime
	smiData.Solid(4).CoM
	mm
	compiletime
	smiData.Solid(4).MoI
	kg*mm^2
	compiletime

Parameter	Value
	smiData.Solid(4).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(4).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(4).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1

Parameter	Value
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J3 (TOP).stp.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.122. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z

Parameter	Value
	smiData.RigidTransform(34).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(34).angle
	compiletime
	smiData.RigidTransform(34).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

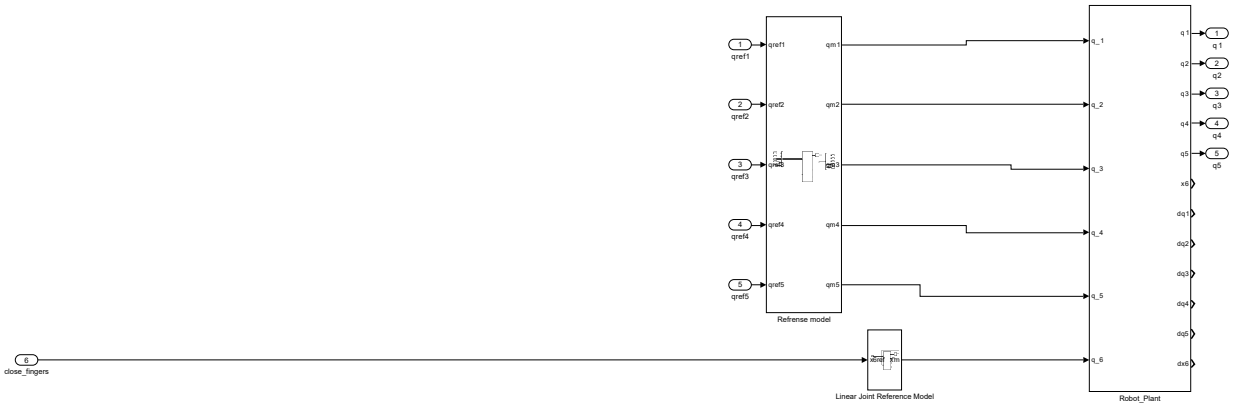
Table 3.123. "Transform1" Parameters

Parameter	Value
	RigidTransform

Parameter	Value
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(37).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(37).angle
	compiletime
	smiData.RigidTransform(37).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

LLC

Figure 3.18. DIY_simuling/LLC



Blocks

Parameters

"close_fingers" (Inport)

Table 3.124. "close_fingers" Parameters

Parameter	Value
Port number	6
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"q1" (Outport)

Table 3.125. "q1" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q2" (Outport)

Table 3.126. "q2" Parameters

Parameter	Value
Port number	2
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off

Parameter	Value
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q3" (Outport)

Table 3.127. "q3" Parameters

Parameter	Value
Port number	3
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q4" (Outport)

Table 3.128. "q4" Parameters

Parameter	Value
Port number	4
Icon display	Port number
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q5" (Outport)

Table 3.129. "q5" Parameters

Parameter	Value
Port number	5
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off

Parameter	Value
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qref1" (Inport)

Table 3.130. "qref1" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"qref2" (Inport)

Table 3.131. "qref2" Parameters

Parameter	Value
Port number	2
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"qref3" (Inport)

Table 3.132. "qref3" Parameters

Parameter	Value
Port number	3
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto

"qref4" (Inport)

Table 3.133. "qref4" Parameters

Parameter	Value
Port number	4
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

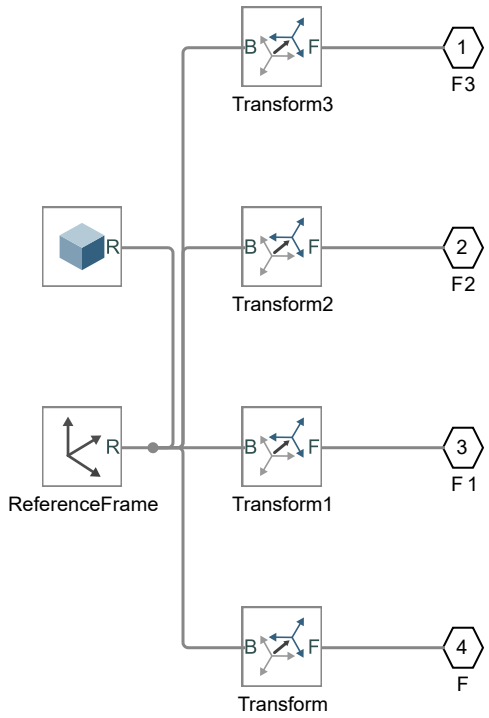
"qref5" (Inport)

Table 3.134. "qref5" Parameters

Parameter	Value
Port number	5
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

main_hand

Figure 3.19. DIY_simuling/LLC/Robot_Plant/EE/main_hand



Blocks

Parameters

"F" (PMIOPort)

Table 3.135. "F" Parameters

Parameter	Value
Port number	4
Port location on parent subsystem	Right

"F1" (PMIOPort)

Table 3.136. "F1" Parameters

Parameter	Value
Port number	3
Port location on parent subsystem	Right

"F2" (PMIOPort)

Table 3.137. "F2" Parameters

Parameter	Value
Port number	2
Port location on parent subsystem	Right

"F3" (PMIOPort)

Table 3.138. "F3" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.139. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.140. "Solid" Parameters

Parameter	Value
	Solid
	Custom
	smiData.Solid(10).mass
	kg

Parameter	Value
	compiletime
	smiData.Solid(10).CoM
	mm
	compiletime
	smiData.Solid(10).MoI
	kg*mm^2
	compiletime
	smiData.Solid(10).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(10).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(10).opacity
	compiletime
	FromFile
	[1 1]
	m
	compiletime

Parameter	Value
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J5 controler.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.141. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(1).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(1).angle
	compiletime
	smiData.RigidTransform(1).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform1" (SimscapeMultibodyBlock)

Table 3.142. "Transform1" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(3).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(3).angle
	compiletime
	smiData.RigidTransform(3).axis
	compiletime
	+X
	+Y
	+Y
	+Z

Parameter	Value
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform2" (SimscapeMultibodyBlock)

Table 3.143. "Transform2" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(5).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(5).angle
	compiletime
	smiData.RigidTransform(5).axis

Parameter	Value
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"Transform3" (SimscapeMultibodyBlock)

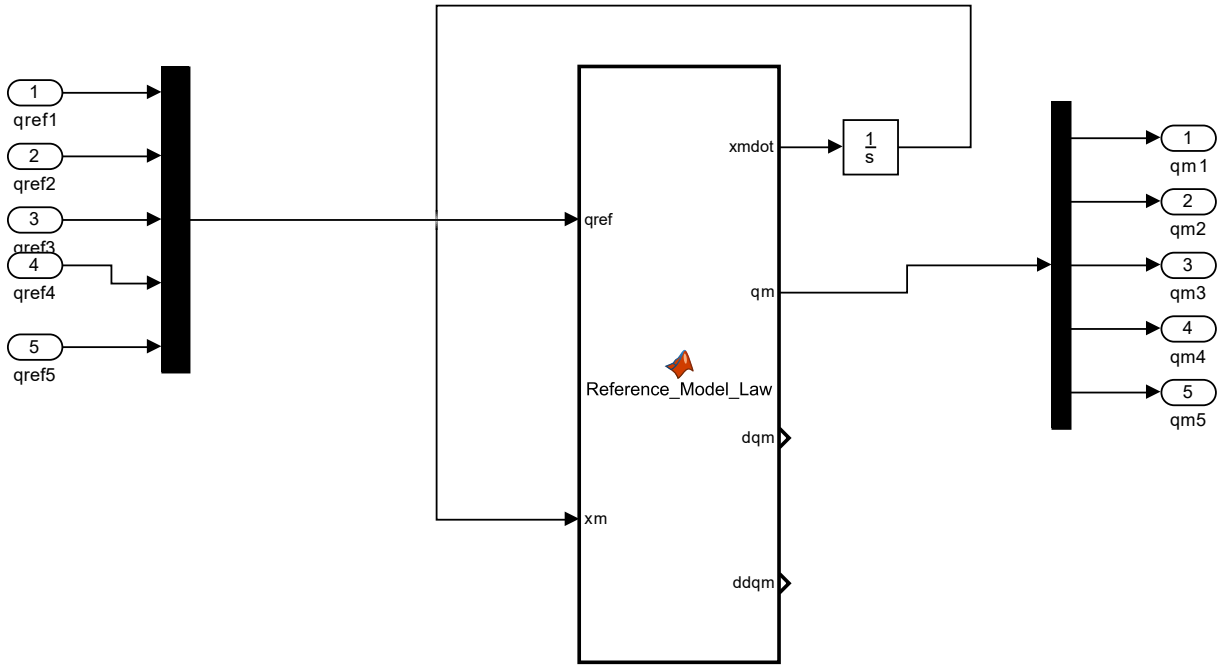
Table 3.144. "Transform3" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(8).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis

Parameter	Value
	rad
	+Z
	smiData.RigidTransform(8).angle
	compiletime
	smiData.RigidTransform(8).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Refrense model

Figure 3.20. DIY_simuling/LLC/Refrense model



Blocks

Parameters

"Demux" (Demux)

Table 3.145. "Demux" Parameters

Parameter	Value
Number of outputs	5
Display option	bar
Bus selection mode	off

"Integrator" (Integrator)

Table 3.146. "Integrator" Parameters

Parameter	Value
External reset	none
Initial condition source	internal
Initial condition	zeros(10,1)
Limit output	off
Upper saturation limit	inf
Lower saturation limit	-inf
Wrap state	off
Wrapped state upper value	pi
Wrapped state lower value	-pi
Show saturation port	off
Show state port	off
Ignore limit and reset when linearizing	off
Enable zero-crossing detection	on
State Name (e.g., 'position')	''

"MATLAB Function" (MATLAB Function)

Table 3.147. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 3.148. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
qref	Input	1	double	5
xmdot	Output	1	double	10
xm	Input	2	double	10
qm	Output	2	double	5
dqm	Output	3	double	5
ddqm	Output	4	double	5

MATLAB Function Function Script

```
function [xmdot, qm, dqm, ddqm] = Reference_Model_Law(qref, xm)

% ============================================================
% Second-order reference model
% qref -> qm, dqm, ddqm
% ============================================================

qref = reshape(qref, [5,1]);
xm   = reshape(xm,   [10,1]);

% Predefine outputs
xmdot = zeros(10,1);
qm    = zeros(5,1);
dqm   = zeros(5,1);
ddqm  = zeros(5,1);

% Reference model parameters
zeta = [1; 1; 1; 1; 1];
wn   = [2; 2; 2; 2; 2];

Z = diag(zeta);
Omega = diag(wn);

Am = [
    zeros(5,5), eye(5);
    -Omega^2,  -2*Z*Omega
];

Bm = [
    zeros(5,5);
    Omega^2
];

% Reference model dynamics
xmdot = Am*xm + Bm*qref;

% Outputs
qm   = xm(1:5);
```

```
dqm  = xm(6:10);
ddqm = xmdot(6:10);

end
```

Table 3.149. MATLAB Function Supporting Functions

Function	Defined By	Path
abs	MATLAB	
diag	MATLAB	
eye	MATLAB	
floor	MATLAB	
full	MATLAB	
intmax	MATLAB	
intmin	MATLAB	
isfi	MATLAB	
isfimath	MATLAB	
isinf	MATLAB	
ismatrix	MATLAB	
isnumerictype	MATLAB	
issparse	MATLAB	
matrix_to_integer_power	MATLAB	
mpower	MATLAB	

"Mux" (Mux)

Table 3.150. "Mux" Parameters

Parameter	Value
Number of inputs	5
Display option	bar

"qm1" (Outport)

Table 3.151. "qm1" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qm2" (Outport)

Table 3.152. "qm2" Parameters

Parameter	Value
Port number	2
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off

Parameter	Value
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qm3" (Outport)

Table 3.153. "qm3" Parameters

Parameter	Value
Port number	3
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qm4" (Outport)

Table 3.154. "qm4" Parameters

Parameter	Value
Port number	4
Icon display	Port number
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qm5" (Outport)

Table 3.155. "qm5" Parameters

Parameter	Value
Port number	5
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off

Parameter	Value
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"qref1" (Inport)

Table 3.156. "qref1" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"qref2" (Inport)

Table 3.157. "qref2" Parameters

Parameter	Value
Port number	2
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"qref3" (Inport)

Table 3.158. "qref3" Parameters

Parameter	Value
Port number	3
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto

"qref4" (Inport)

Table 3.159. "qref4" Parameters

Parameter	Value
Port number	4
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

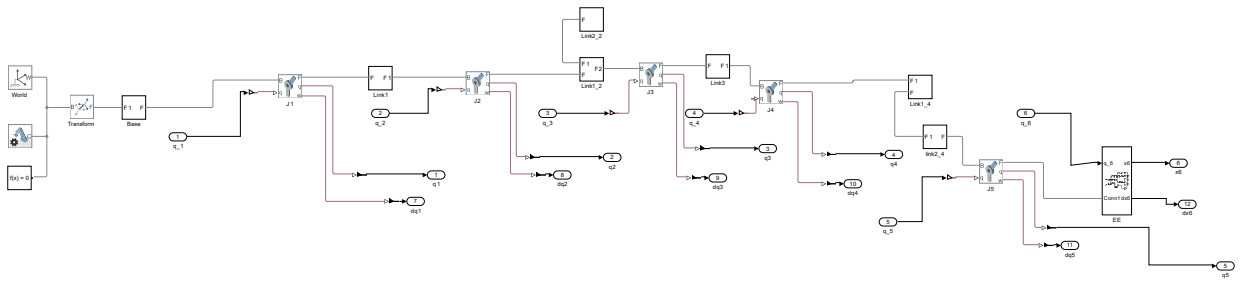
"qref5" (Inport)

Table 3.160. "qref5" Parameters

Parameter	Value
Port number	5
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

Robot_Plant

Figure 3.21. DIY_simuling/LLC/Robot_Plant



Blocks

Parameters

"dq1" (Outport)

Table 3.161. "dq1" Parameters

Parameter	Value
Port number	7
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"dq2" (Outport)

Table 3.162. "dq2" Parameters

Parameter	Value
Port number	8
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto

Parameter	Value
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"dq3" (Outport)

Table 3.163. "dq3" Parameters

Parameter	Value
Port number	9
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off

Parameter	Value
Constant value	0
Interpret vector parameters as 1-D	off

"dq4" (Outport)

Table 3.164. "dq4" Parameters

Parameter	Value
Port number	10
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"dq5" (Outport)

Table 3.165. "dq5" Parameters

Parameter	Value
Port number	11
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto

Parameter	Value
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"dx6" (Outport)

Table 3.166. "dx6" Parameters

Parameter	Value
Port number	12
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off

Parameter	Value
Constant value	0
Interpret vector parameters as 1-D	off

"J1" (SimscapeMultibodyBlock)

Table 3.167. "J1" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	ComputedTorque

Parameter	Value
	InputMotion
	on
	on
	off
	off
	off
	off
	on
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(5).Rz.Pos
	deg

Parameter	Value
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"J2" (SimscapeMultibodyBlock)

Table 3.168. "J2" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad

Parameter	Value
	compiletime
	0
	N*m/(rad/s)
	compiletime
	ComputedTorque
	InputMotion
	on
	on
	off
	off
	off
	off
	on
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg

Parameter	Value
	compiletime
	on
	Low
	smiData.RevoluteJoint(6).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"J3" (SimscapeMultibodyBlock)

Table 3.169. "J3" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame

Parameter	Value
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	ComputedTorque
	InputMotion
	on
	on
	off
	off
	off
	off
	on
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime

Parameter	Value
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(13).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"J4" (SimscapeMultibodyBlock)

Table 3.170. "J4" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off
	off

Parameter	Value
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	ComputedTorque
	InputMotion
	on
	on
	off
	off
	off
	off
	on
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	90

Parameter	Value
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(14).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"J5" (SimscapeMultibodyBlock)

Table 3.171. "J5" Parameters

Parameter	Value
	RevoluteJoint
	off
	off
	off
	off
	off
	off
	off
	off

Parameter	Value
	off
	off
	off
	off
	off
	off
	off
	off
	FollowerOnBase
	BaseFrame
	0
	rad
	compiletime
	0
	N*m/rad
	compiletime
	0
	N*m/(rad/s)
	compiletime
	ComputedTorque
	InputMotion
	on
	on
	off
	off
	off
	off
	on
	-90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime

Parameter	Value
	0.1
	deg
	compiletime
	on
	90
	deg
	compiletime
	1e4
	N*m/deg
	compiletime
	10
	N*m/(deg/s)
	compiletime
	0.1
	deg
	compiletime
	on
	Low
	smiData.RevoluteJoint(4).Rz.Pos
	deg
	compiletime
	off
	High
	0
	deg/s
	compiletime
	simmechanics.library.joints.revolute_joint

"MechanismConfiguration" (SimscapeMultibodyBlock)

Table 3.172. "MechanismConfiguration" Parameters

Parameter	Value
	MechanismConfiguration
	Constant
	[0 -9.80665 0]
	m/s^2

Parameter	Value
	compiletime
	0.001
	simmechanics.library.utilities.mechanism_configuration

"PS-Simulink Converter" (SubSystem)

Table 3.173. "PS-Simulink Converter" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad
AffineConversion	off

"PS-Simulink Converter1" (SubSystem)

Table 3.174. "PS-Simulink Converter1" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad
AffineConversion	off

"PS-Simulink Converter2" (SubSystem)

Table 3.175. "PS-Simulink Converter2" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output

Parameter	Value
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad
AffineConversion	off

"PS-Simulink Converter3" (SubSystem)

Table 3.176. "PS-Simulink Converter3" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad
AffineConversion	off

"PS-Simulink Converter4" (SubSystem)

Table 3.177. "PS-Simulink Converter4" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad
AffineConversion	off

"PS-Simulink Converter5" (SubSystem)

Table 3.178. "PS-Simulink Converter5" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad/s
AffineConversion	off

"PS-Simulink Converter6" (SubSystem)

Table 3.179. "PS-Simulink Converter6" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad/s
AffineConversion	off

"PS-Simulink Converter7" (SubSystem)

Table 3.180. "PS-Simulink Converter7" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]

Parameter	Value
Unit	rad/s
AffineConversion	off

"PS-Simulink Converter8" (SubSystem)

Table 3.181. "PS-Simulink Converter8" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad/s
AffineConversion	off

"PS-Simulink Converter9" (SubSystem)

Table 3.182. "PS-Simulink Converter9" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_output
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
Unit	rad/s
AffineConversion	off

"q1" (Outport)

Table 3.183. "q1" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Minimum	[]

Parameter	Value
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q2" (Outport)

Table 3.184. "q2" Parameters

Parameter	Value
Port number	2
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]

Parameter	Value
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q3" (Outport)

Table 3.185. "q3" Parameters

Parameter	Value
Port number	3
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q4" (Outport)

Table 3.186. "q4" Parameters

Parameter	Value
Port number	4
Icon display	Port number
Minimum	[]

Parameter	Value
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q5" (Outport)

Table 3.187. "q5" Parameters

Parameter	Value
Port number	5
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]

Parameter	Value
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"q_1" (Inport)

Table 3.188. "q_1" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"q_2" (Inport)

Table 3.189. "q_2" Parameters

Parameter	Value
Port number	2
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"q_3" (Inport)

Table 3.190. "q_3" Parameters

Parameter	Value
Port number	3
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]

Parameter	Value
Data type	Inherit: auto

"q_4" (Inport)

Table 3.191. "q_4" Parameters

Parameter	Value
Port number	4
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"q_5" (Inport)

Table 3.192. "q_5" Parameters

Parameter	Value
Port number	5
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"q_6" (Inport)

Table 3.193. "q_6" Parameters

Parameter	Value
Port number	6
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"Simulink-PS Converter1" (SubSystem)

Table 3.194. "Simulink-PS Converter1" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	rad
AffineConversion	off
FilteringAndDerivatives	filter
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001
UdotUserProvided	0

"Simulink-PS Converter2" (SubSystem)

Table 3.195. "Simulink-PS Converter2" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	rad
AffineConversion	off
FilteringAndDerivatives	filter
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001

Parameter	Value
UdotUserProvided	0

"Simulink-PS Converter3" (SubSystem)

Table 3.196. "Simulink-PS Converter3" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	rad
AffineConversion	off
FilteringAndDerivatives	filter
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001
UdotUserProvided	0

"Simulink-PS Converter4" (SubSystem)

Table 3.197. "Simulink-PS Converter4" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	rad
AffineConversion	off
FilteringAndDerivatives	filter

Parameter	Value
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001
UdotUserProvided	0

"Simulink-PS Converter5" (SubSystem)

Table 3.198. "Simulink-PS Converter5" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	ps_input
LeftPortType	input
RightPortType	output
PseudoPeriodic	off
Frequencies	[]
NoiseDistribution	none
NoiseParameters	[]
Unit	rad
AffineConversion	off
FilteringAndDerivatives	filter
SimscapeFilterOrder	2
InputFilterTimeConstant	0.001
UdotUserProvided	0

"Solver Configuration" (SubSystem)

Table 3.199. "Solver Configuration" Parameters

Parameter	Value
PhysicalDomain	network_engine_domain
SubClassName	solver
LeftPortType	input
RightPortType	generic
Tones	[]
Harmonics	[]
SimulateNoise	off
Profile	off
UseLocalSampling	off

Parameter	Value
EnableSwitchedLinearOptims	on
FrequencyDomain	off
UseCCode	off
RelTol	0.001
AbsTol	1e-06
MinStep	1e-09
ToWorkspace	off
RevertToSquareIcSolve	off
NumHomotopyIterations	100
PhasorMode	off
PhasorModeNumHarmonics	1
DoDC	off
ResidualTolerance	1e-09
UseLocalSolver	off
LocalSolverChoice	NE_BACKWARD_EULER_ADVANCER
LocalSolverSampleTime	0.001
DoFixedCost	off
MaxNonlinIter	3
MaxModeIter	2
EquationFormulation	NE_TIME_EF
DelaysMemoryBudget	1024
AutomaticFiltering	on
FilteringTimeConstant	0.001
PartitionStorageMethod	AS_NEEDED
PartitionMemoryBudget	1024
PartitionMethod	ROBUST

"Transform" (SimscapeMultibodyBlock)

Table 3.200. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime

Parameter	Value
	+Z
	smiData.RigidTransform(39).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	RotationSequence
	rad
	+Z
	smiData.RigidTransform(39).angle
	compiletime
	smiData.RigidTransform(39).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	BaseAxes
	XYZ
	[90 0 90]
	deg
	compiletime
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

"World" (SimscapeMultibodyBlock)

Table 3.201. "World" Parameters

Parameter	Value
	WorldFrame
	simmechanics.library.frames_transforms.world_frame

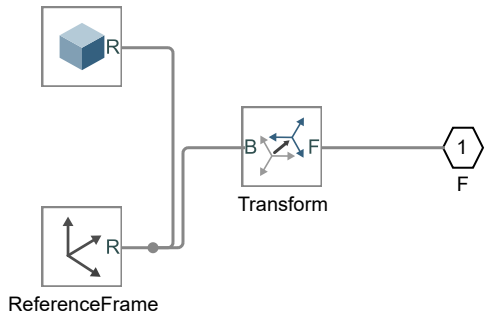
"x6" (Outport)

Table 3.202. "x6" Parameters

Parameter	Value
Port number	6
Icon display	Port number
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Source of initial output value	Dialog
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

scrow

Figure 3.22. DIY_simuling/LLC/Robot_Plant/EE/scrow



Blocks

Parameters

"F" (PMIOPort)

Table 3.203. "F" Parameters

Parameter	Value
Port number	1
Port location on parent subsystem	Left

"ReferenceFrame" (SimscapeMultibodyBlock)

Table 3.204. "ReferenceFrame" Parameters

Parameter	Value
	ReferenceFrame
	simmechanics.library.frames_transforms.reference_frame

"Solid" (SimscapeMultibodyBlock)

Table 3.205. "Solid" Parameters

Parameter	Value
	Solid
	Custom

Parameter	Value
	smiData.Solid(9).mass
	kg
	compiletime
	smiData.Solid(9).CoM
	mm
	compiletime
	smiData.Solid(9).MoI
	kg*mm^2
	compiletime
	smiData.Solid(9).PoI
	kg*mm^2
	compiletime
	Density
	1000
	kg/m^3
	compiletime
	on
	<Frames/>
	FromGeometry
	Sphere
	10
	compiletime
	SimpleVisualProperties
	smiData.Solid(9).color
	compiletime
	[0.5 0.5 0.5 1.0]
	compiletime
	[0.15 0.15 0.15 1.0]
	compiletime
	[0.0 0.0 0.0 1.0]
	compiletime
	75
	compiletime
	smiData.Solid(9).opacity
	compiletime
	FromFile
	[1 1]

Parameter	Value
	m
	compiletime
	[1 1 1]
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	1
	m
	compiletime
	[1 1 1]
	m
	compiletime
	[1 1; -1 1; -1 -1; 1 -1]
	m
	1
	m
	3
	1
	m
	compiletime
	STEP
	Robot - J7 thread.step_Default_sldprt.STEP
	m
	[1 1; 1 -1; 2 -1; 2 1]
	m
	Full
	180
	deg
	simmechanics.library.body_elements.solid

"Transform" (SimscapeMultibodyBlock)

Table 3.206. "Transform" Parameters

Parameter	Value
	RigidTransform
	Cartesian
	0
	mm
	compiletime
	+Z
	smiData.RigidTransform(4).translation
	compiletime
	0
	m
	compiletime
	0
	m
	compiletime
	0
	deg
	compiletime
	ArbitraryAxis
	rad
	+Z
	smiData.RigidTransform(4).angle
	compiletime
	smiData.RigidTransform(4).axis
	compiletime
	+X
	+Y
	+Y
	+Z
	FollowerAxes
	XYX
	[0 0 0]
	deg
	compiletime

Parameter	Value
	[1 0 0; 0 1 0; 0 0 1]
	compiletime
	simmechanics.library.frames_transforms.rigid_transform

Chapter 4. System Design Variables

Design Variable Summary

Table 4.1. Design Variables

Variable Name	Parent Blocks	Size	Bytes	Class	Value
smiData	Transform Transform Transform1 Transform Transform1 Transform Transform1 Transform Transform Transform1 Transform2 Transform Transform Transform Transform1 Transform3 Transform Transform Transform1 Transform1 Transform1 Transform2 Transform1 Transform1 Transform3 Transform1 Transform2 Transform Transform Transform1 Transform1 Transform1 Transform Transform Transform2 Transform Transform Transform3 Transform	1x1	47646	struct	RigidTransform: [1×39 struct] Solid: [1×13 struct] CylindricalJoint: [1×3 struct] RevoluteJoint: [1×14 struct]

Variable Name	Parent Blocks	Size	Bytes	Class	Value
	Solid Solid Solid

Variable Name	Parent Blocks	Size	Bytes	Class	Value
	Solid Transform Solid Transform Transform1 Solid Transform Transform1 Solid Transform Transform1 Solid Transform Transform1 Solid Transform Transform1 Solid Transform Transform1 Solid Transform Transform1 Transform2 Transform3 Revolute1 Revolute10 Revolute11 Revolute6 Revolute7 Revolute8 Revolute9 Solid Transform Transform1 Transform2 Transform3 Solid Transform Transform1 Transform2 Transform3 Solid Transform J1 J2 J3 J4 J5 Solid Transform				

Variable Name	Parent Blocks	Size	Bytes	Class	Value
	Transform1 Solid Transform Transform1 Transform2 Solid Transform Transform1 Solid Transform Solid Transform Transform1 Transform Solid Transform Transform1				

Table 4.2. Functions used in Design Variable Expressions

Function Name	Parent Blocks	Calling character vector
pi	Integrator Integrator Integrator Integrator	-pi -pi pi pi
zeros	Integrator	zeros(10,1)

Design Variable Details

Table 4.3. smiData

Field	Value
RigidTransform	[smiData.RigidTransform(1), smiData.RigidTransform(2), smiData.RigidTransform(3), smiData.RigidTransform(4), smiData.RigidTransform(5), smiData.RigidTransform(6), smiData.RigidTransform(7), smiData.RigidTransform(8), smiData.RigidTransform(9), smiData.RigidTransform(10), smiData.RigidTransform(11), smiData.RigidTransform(12), smiData.RigidTransform(13), smiData.RigidTransform(14), smiData.RigidTransform(15), smiData.RigidTransform(16), smiData.RigidTransform(17), smiData.RigidTransform(18), smiData.RigidTransform(19), smiData.RigidTransform(20), smiData.RigidTransform(21), smiData.RigidTransform(22), smiData.RigidTransform(23), smiData.RigidTransform(24), smiData.RigidTransform(25), smiData.RigidTransform(26), smiData.RigidTransform(27), smiData.RigidTransform(28), smiData.RigidTransform(29), smiData.RigidTransform(30), smiData.RigidTransform(31), smiData.RigidTransform(32), smiData.RigidTransform(33), smiData.RigidTransform(34), smiData.RigidTransform(35), smiData.RigidTr

	ansform(36), smiData.RigidTransform(37), smiData.RigidTransform(38), smiData.RigidTransform(39)]
Solid	[smiData.Solid(1), smiData.Solid(2), smiData.Solid(3), smiData.Solid(4), smiData.Solid(5), smiData.Solid(6), smiData.Solid(7), smiData.Solid(8), smiData.Solid(9), smiData.Solid(10), smiData.Solid(11), smiData.Solid(12), smiData.Solid(13)]
CylindricalJoint	[smiData.CylindricalJoint(1), smiData.CylindricalJoint(2), smiData.CylindricalJoint(3)]
RevoluteJoint	[smiData.RevoluteJoint(1), smiData.RevoluteJoint(2), smiData.RevoluteJoint(3), smiData.RevoluteJoint(4), smiData.RevoluteJoint(5), smiData.RevoluteJoint(6), smiData.RevoluteJoint(7), smiData.RevoluteJoint(8), smiData.RevoluteJoint(9), smiData.RevoluteJoint(10), smiData.RevoluteJoint(11), smiData.RevoluteJoint(12), smiData.RevoluteJoint(13), smiData.RevoluteJoint(14)]

Table 4.4. smiData.RigidTransform(1)

Field	Value
translation	[0 0 48]
angle	3.1416
axis	[1 0 0]
ID	B[Robot - J5 controler.step-1:-:Robot - J6 hand.step-1]

Table 4.5. smiData.RigidTransform(2)

Field	Value
translation	[-9.85112e-12 -1.76259e-12 41.7417]
angle	3.1416
axis	[1 7.9442e-33 4.9369e-17]
ID	F[Robot - J5 controler.step-1:-:Robot - J6 hand.step-1]

Table 4.6. smiData.RigidTransform(3)

Field	Value
translation	[0 0 48]
angle	3.1416
axis	[1 0 0]
ID	B[Robot - J5 controler.step-1:-:Robot - J7 thread.step-1]

Table 4.7. smiData.RigidTransform(4)

Field	Value
translation	[6.0818e-13 1.00044e-11 43]

angle	3.1416
axis	[-1 8.699e-33 -3.7102e-17]
ID	F[Robot - J5 controler.step-1:-:Robot - J7 thread.step-1]

Table 4.8. smiData.RigidTransform(5)

Field	Value
translation	[0 0 48]
angle	3.1416
axis	[1 0 0]
ID	B[Robot - J5 controler.step-1:-:Robot - J8 cap.step-1]

Table 4.9. smiData.RigidTransform(6)

Field	Value
translation	[4.5589e-12 1.235e-11 9]
angle	3.1416
axis	[1 8.1716e-33 1.9569e-16]
ID	F[Robot - J5 controler.step-1:-:Robot - J8 cap.step-1]

Table 4.10. smiData.RigidTransform(7)

Field	Value
translation	[26 0 150]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J4 (J6 JOINT).stp.step-1:-:Robot - J5 controler.step-1]

Table 4.11. smiData.RigidTransform(8)

Field	Value
translation	[25.4 1.88738e-14 -38]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J4 (J6 JOINT).stp.step-1:-:Robot - J5 controler.step-1]

Table 4.12. smiData.RigidTransform(9)

Field	Value

translation	[0 60 0]
angle	2.0944
axis	[0.57735 -0.57735 0.57735]
ID	B[Robot - Base.step-1:-:Robot - J1.stp.step-1]

Table 4.13. smiData.RigidTransform(10)

Field	Value
translation	[-7.1047e-15 0 -9.8253e-17]
angle	2.0944
axis	[0.57735 -0.57735 0.57735]
ID	F[Robot - Base.step-1:-:Robot - J1.stp.step-1]

Table 4.14. smiData.RigidTransform(11)

Field	Value
translation	[-25 50.0544 17.4148]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J1.stp.step-1:-:Robot - J2 (LEFT).stp.step-1]

Table 4.15. smiData.RigidTransform(12)

Field	Value
translation	[-27.5 7.01696e-16 5.55112e-15]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J1.stp.step-1:-:Robot - J2 (LEFT).stp.step-1]

Table 4.16. smiData.RigidTransform(13)

Field	Value
translation	[-10.5 142.1868 0]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	B[Robot - J2 (LEFT).stp.step-1:-:Robot - J2 (RIGHT).stp.step-1]

Table 4.17. smiData.RigidTransform(14)

Field	Value
translation	[-10.5 142.1868 8.828493e-13]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	F[Robot - J2 (LEFT).stp.step-1:-:Robot - J2 (RIGHT).stp.step-1]

Table 4.18. smiData.RigidTransform(15)

Field	Value
translation	[-46.1502 48.6349 30.2747]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	B[Robot - J6 hand.step-1:-:Robot - J9 hanger-1]

Table 4.19. smiData.RigidTransform(16)

Field	Value
translation	[50.6502 1.82252e-11 11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J6 hand.step-1:-:Robot - J9 hanger-1]

Table 4.20. smiData.RigidTransform(17)

Field	Value
translation	[4 15 -25.9808]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J10 finger.step-1:-:Robot - J9 hanger-1]

Table 4.21. smiData.RigidTransform(18)

Field	Value
translation	[6.5 -1.14027e-11 -11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]

ID	F[Robot - J10 finger.step-1:-:Robot - J9 hanger-1]

Table 4.22. smiData.RigidTransform(19)

Field	Value
translation	[76.2317 34.7673 30.2747]
angle	1.8235
axis	[0.44721 -0.7746 0.44721]
ID	B[Robot - J6 hand.step-1:-:Robot - J9 hanger-2]

Table 4.23. smiData.RigidTransform(20)

Field	Value
translation	[72.7252 4.14246e-11 11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J6 hand.step-1:-:Robot - J9 hanger-2]

Table 4.24. smiData.RigidTransform(21)

Field	Value
translation	[4 15 -25.9808]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J10 finger.step-2:-:Robot - J9 hanger-2]

Table 4.25. smiData.RigidTransform(22)

Field	Value
translation	[6.5 1.90159e-12 -11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J10 finger.step-2:-:Robot - J9 hanger-2]

Table 4.26. smiData.RigidTransform(23)

Field	Value
translation	[-18.544 -65.1507 30.2747]
angle	1.8235

axis	[-0.44721 -0.7746 -0.44721]
ID	B[Robot - J6 hand.step-1:-:Robot - J9 hanger-3]

Table 4.27. smiData.RigidTransform(24)

Field	Value
translation	[51.6502 8.95284e-13 11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J6 hand.step-1:-:Robot - J9 hanger-3]

Table 4.28. smiData.RigidTransform(25)

Field	Value
translation	[4 15 -25.9808]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J10 finger.step-3:-:Robot - J9 hanger-3]

Table 4.29. smiData.RigidTransform(26)

Field	Value
translation	[6.5 -4.42544e-11 -11.5]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J10 finger.step-3:-:Robot - J9 hanger-3]

Table 4.30. smiData.RigidTransform(27)

Field	Value
translation	[19.9266 -3.65269 7.46741]
angle	1.8235
axis	[0.44721 -0.7746 0.44721]
ID	B[Robot - J8 cap.step-1:-:Robot - J10 finger.step-1]

Table 4.31. smiData.RigidTransform(28)

Field	Value
translation	[8.8 7.3973e-11 3.8857e-11]

angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J8 cap.step-1:-:Robot - J10 finger.step-1]

Table 4.32. smiData.RigidTransform(29)

Field	Value
translation	[-13.1266 -15.4306 7.46741]
angle	1.8235
axis	[-0.44721 -0.7746 -0.44721]
ID	B[Robot - J8 cap.step-1:-:Robot - J10 finger.step-2]

Table 4.33. smiData.RigidTransform(30)

Field	Value
translation	[8.8 8.6011e-12 -5.4512e-11]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J8 cap.step-1:-:Robot - J10 finger.step-2]

Table 4.34. smiData.RigidTransform(31)

Field	Value
translation	[-6.8 19.0833 7.46741]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	B[Robot - J8 cap.step-1:-:Robot - J10 finger.step-3]

Table 4.35. smiData.RigidTransform(32)

Field	Value
translation	[8.8 2.2148e-11 -3.9968e-11]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J8 cap.step-1:-:Robot - J10 finger.step-3]

Table 4.36. smiData.RigidTransform(33)

Field	Value

translation	[-10.5 120 0]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	B[Robot - J2 (LEFT).stp.step-1:-:Robot - J3 (TOP).stp.step-1]

Table 4.37. smiData.RigidTransform(34)

Field	Value
translation	[-7.59 -42.6255 -20]
angle	2.0944
axis	[-0.57735 -0.57735 0.57735]
ID	F[Robot - J2 (LEFT).stp.step-1:-:Robot - J3 (TOP).stp.step-1]

Table 4.38. smiData.RigidTransform(35)

Field	Value
translation	[0 14 96]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	B[Robot - J4 (ARM).stp.step-1:-:Robot - J4 (J6 JOINT).stp.step-1]

Table 4.39. smiData.RigidTransform(36)

Field	Value
translation	[4.06342e-14 14 96]
angle	2.0944
axis	[0.57735 0.57735 0.57735]
ID	F[Robot - J4 (ARM).stp.step-1:-:Robot - J4 (J6 JOINT).stp.step-1]

Table 4.40. smiData.RigidTransform(37)

Field	Value
translation	[0 0 0]
angle	0
axis	[0 0 0]
ID	B[Robot - J3 (TOP).stp.step-1:-:Robot - J4 (ARM).stp.step-1]

Table 4.41. smiData.RigidTransform(38)

Field	Value
translation	[9.8368e-12 1.9333e-12 7.1937e-14]
angle	1.4249e-16
axis	[-0.68369 0.72978 -3.5548e-17]
ID	F[Robot - J3 (TOP).stp.step-1:-:Robot - J4 (ARM).stp.step-1]

Table 4.42. smiData.RigidTransform(39)

Field	Value
translation	[58.9038 35.6268 90.2376]
angle	0
axis	[0 0 0]
ID	RootGround[Robot - Base.step-1]

Table 4.43. smiData.Solid(1)

Field	Value
mass	0.2562
CoM	[-5.26441e-05 28.7431 0.62099]
MoI	[371.4564 587.2715 377.3818]
PoI	[-2.0842 8.2729e-05 -0.00047508]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - Base.step*:*Default

Table 4.44. smiData.Solid(2)

Field	Value
mass	0.2078
CoM	[-2.64034 15.7314 -2.94975]
MoI	[167.4002 240.4168 229.4996]
PoI	[-3.68504 3.34098 13.8406]
color	[0.96078 0.80392 0]
opacity	1
ID	Robot - J1.stp.step*:*Default

Table 4.45. smiData.Solid(3)

Field	Value
mass	0.0666
CoM	[-17.5786 44.1525 -5.6114e-07]
MoI	[147.9945 8.912998 141.8995]
PoI	[-9.5694e-07 1.8353e-07 -2.9333]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J2 (LEFT).stp.step*:*Default

Table 4.46. smiData.Solid(4)

Field	Value
mass	0.0719
CoM	[2.12245 -5.40144 -25.604]
MoI	[53.6929 42.2445 41.8961]
PoI	[3.3511 1.2939 -1.7874]
color	[0.96078 0.80392 0]
opacity	1
ID	Robot - J3 (TOP).stp.step*:*Default

Table 4.47. smiData.Solid(5)

Field	Value
mass	0.0693
CoM	[11.4579 37.946 -4.95861e-06]
MoI	[93.0572 13.3635 90.9544]
PoI	[0.00019555 2.5061e-05 2.4055]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J2 (RIGHT).stp.step*:*Default

Table 4.48. smiData.Solid(6)

Field	Value
mass	0.1290
CoM	[-3.75509 2.38797 57.3689]

MoI	[235.0791 264.0093 48.69327]
PoI	[6.99772 35.4167 -1.15689]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J4 (ARM).stp.step*:*Default

Table 4.49. smiData.Solid(7)

Field	Value
mass	0.0028
CoM	[0.026326 -0.061684 6.0178]
MoI	[0.28484 0.28465 0.51789]
PoI	[-0.00028577 6.3046e-05 0.00010264]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J8 cap.step*:*Default

Table 4.50. smiData.Solid(8)

Field	Value
mass	8.3915e-04
CoM	[4.5 0 0]
MoI	[0.039099 0.044713 0.010755]
PoI	[0 0 0]
color	[0.7 0.7 0.7]
opacity	1
ID	Robot - J9 hanger*:*Default

Table 4.51. smiData.Solid(9)

Field	Value
mass	0.0026
CoM	[-0.0177813 0.0255354 21.2841]
MoI	[0.37986 0.37987 0.028705]
PoI	[9.5911e-05 -0.00027137 4.6262e-06]
color	[0.79216 0.81961 0.93333]
opacity	1
ID	Robot - J7 thread.step*:*Default

Table 4.52. smiData.Solid(10)

Field	Value
mass	0.0507
CoM	[0.0161112 0.166976 -12.8834]
MoI	[36.1825 39.2715 13.6817]
PoI	[0.19089 0.021468 0.15822]
color	[0.96078 0.80392 0]
opacity	1
ID	Robot - J5 controler.step*:*Default

Table 4.53. smiData.Solid(11)

Field	Value
mass	0.0091
CoM	[-0.00247708 0.00288955 10.3884]
MoI	[3.4433 3.4433 5.5316]
PoI	[0.00040619 0.0001686 2.1257e-05]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J6 hand.step*:*Default

Table 4.54. smiData.Solid(12)

Field	Value
mass	0.0023
CoM	[2 4.46956 -30.3062]
MoI	[0.9193 0.81554 0.10966]
PoI	[-0.1387 -7.0269e-08 -3.7992e-08]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J10 finger.step*:*Default

Table 4.55. smiData.Solid(13)

Field	Value
mass	0.0264
CoM	[19.51069 -0.0001016129 123.7889]

MoI	[12.4157 12.1749 3.77142]
PoI	[-7.9153e-06 -3.4826 3.301e-05]
color	[0.29804 0.29804 0.29804]
opacity	1
ID	Robot - J4 (J6 JOINT).stp.step*:*Default

Table 4.56. smiData.CylindricalJoint(1)

Field	Value
Rz	smiData.CylindricalJoint(1).Rz
Pz	smiData.CylindricalJoint(1).Pz
ID	[Robot - J8 cap.step-1:-:Robot - J10 finger.step-1]

Table 4.57. smiData.CylindricalJoint(2)

Field	Value
Rz	smiData.CylindricalJoint(2).Rz
Pz	smiData.CylindricalJoint(2).Pz
ID	[Robot - J8 cap.step-1:-:Robot - J10 finger.step-2]

Table 4.58. smiData.CylindricalJoint(3)

Field	Value
Rz	smiData.CylindricalJoint(3).Rz
Pz	smiData.CylindricalJoint(3).Pz
ID	[Robot - J8 cap.step-1:-:Robot - J10 finger.step-3]

Table 4.59. smiData.RevoluteJoint(1)

Field	Value
Rz	smiData.RevoluteJoint(1).Rz
ID	[Robot - J5 controler.step-1:-:Robot - J6 hand.step-1]

Table 4.60. smiData.RevoluteJoint(2)

Field	Value
Rz	smiData.RevoluteJoint(2).Rz
ID	[Robot - J5 controler.step-1:-:Robot - J7 thread.step-1]

Table 4.61. smiData.RevoluteJoint(3)

Field	Value
Rz	smiData.RevoluteJoint(3).Rz
ID	[Robot - J5 controler.step-1:-:Robot - J8 cap.step-1]

Table 4.62. smiData.RevoluteJoint(4)

Field	Value
Rz	smiData.RevoluteJoint(4).Rz
ID	[Robot - J4 (J6 JOINT).stp.step-1:-:Robot - J5 controler.step-1]

Table 4.63. smiData.RevoluteJoint(5)

Field	Value
Rz	smiData.RevoluteJoint(5).Rz
ID	[Robot - Base.step-1:-:Robot - J1.stp.step-1]

Table 4.64. smiData.RevoluteJoint(6)

Field	Value
Rz	smiData.RevoluteJoint(6).Rz
ID	[Robot - J1.stp.step-1:-:Robot - J2 (LEFT).stp.step-1]

Table 4.65. smiData.RevoluteJoint(7)

Field	Value
Rz	smiData.RevoluteJoint(7).Rz
ID	[Robot - J6 hand.step-1:-:Robot - J9 hanger-1]

Table 4.66. smiData.RevoluteJoint(8)

Field	Value
Rz	smiData.RevoluteJoint(8).Rz
ID	[Robot - J10 finger.step-1:-:Robot - J9 hanger-1]

Table 4.67. smiData.RevoluteJoint(9)

Field	Value

Rz	smiData.RevoluteJoint(9).Rz
ID	[Robot - J6 hand.step-1:-:Robot - J9 hanger-2]

Table 4.68. smiData.RevoluteJoint(10)

Field	Value
Rz	smiData.RevoluteJoint(10).Rz
ID	[Robot - J10 finger.step-2:-:Robot - J9 hanger-2]

Table 4.69. smiData.RevoluteJoint(11)

Field	Value
Rz	smiData.RevoluteJoint(11).Rz
ID	[Robot - J6 hand.step-1:-:Robot - J9 hanger-3]

Table 4.70. smiData.RevoluteJoint(12)

Field	Value
Rz	smiData.RevoluteJoint(12).Rz
ID	[Robot - J10 finger.step-3:-:Robot - J9 hanger-3]

Table 4.71. smiData.RevoluteJoint(13)

Field	Value
Rz	smiData.RevoluteJoint(13).Rz
ID	[Robot - J2 (LEFT).stp.step-1:-:Robot - J3 (TOP).stp.step-1]

Table 4.72. smiData.RevoluteJoint(14)

Field	Value
Rz	smiData.RevoluteJoint(14).Rz
ID	[Robot - J3 (TOP).stp.step-1:-:Robot - J4 (ARM).stp.step-1]

Table 4.73. smiData.CylindricalJoint(1).Rz

Field	Value
Pos	-124.4663

Table 4.74. smiData.CylindricalJoint(1).Pz

Field	Value
Pos	0

Table 4.75. smiData.CylindricalJoint(2).Rz

Field	Value
Pos	-124.4663

Table 4.76. smiData.CylindricalJoint(2).Pz

Field	Value
Pos	0

Table 4.77. smiData.CylindricalJoint(3).Rz

Field	Value
Pos	-34.4663

Table 4.78. smiData.CylindricalJoint(3).Pz

Field	Value
Pos	0

Table 4.79. smiData.RevoluteJoint(1).Rz

Field	Value
Pos	-0.6685

Table 4.80. smiData.RevoluteJoint(2).Rz

Field	Value
Pos	-104.7591

Table 4.81. smiData.RevoluteJoint(3).Rz

Field	Value
Pos	-120.6685

Table 4.82. smiData.RevoluteJoint(4).Rz

Field	Value
Pos	-0.1922

Table 4.83. smiData.RevoluteJoint(5).Rz

Field	Value
Pos	179.2077

Table 4.84. smiData.RevoluteJoint(6).Rz

Field	Value
Pos	-1.8442

Table 4.85. smiData.RevoluteJoint(7).Rz

Field	Value
Pos	6.1941

Table 4.86. smiData.RevoluteJoint(8).Rz

Field	Value
Pos	40.6604

Table 4.87. smiData.RevoluteJoint(9).Rz

Field	Value
Pos	-83.8059

Table 4.88. smiData.RevoluteJoint(10).Rz

Field	Value
Pos	40.6604

Table 4.89. smiData.RevoluteJoint(11).Rz

Field	Value
Pos	-83.8059

Table 4.90. smiData.RevoluteJoint(12).Rz

Field	Value
Pos	40.6604

Table 4.91. smiData.RevoluteJoint(13).Rz

Field	Value
Pos	-0.7650

Table 4.92. smiData.RevoluteJoint(14).Rz

Field	Value
Pos	-0.3040

Used by Blocks:

- DIY_simuling/LLC/Robot_Plant/Base/Solid
- DIY_simuling/LLC/Robot_Plant/Base/Transform
- DIY_simuling/LLC/Robot_Plant/EE/F1/Solid
- DIY_simuling/LLC/Robot_Plant/EE/F1/Transform
- DIY_simuling/LLC/Robot_Plant/EE/F1/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/F2/Solid
- DIY_simuling/LLC/Robot_Plant/EE/F2/Transform
- DIY_simuling/LLC/Robot_Plant/EE/F2/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/F3/Solid
- DIY_simuling/LLC/Robot_Plant/EE/F3/Transform
- DIY_simuling/LLC/Robot_Plant/EE/F3/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/H1/Solid
- DIY_simuling/LLC/Robot_Plant/EE/H1/Transform
- DIY_simuling/LLC/Robot_Plant/EE/H1/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/H2/Solid
- DIY_simuling/LLC/Robot_Plant/EE/H2/Transform
- DIY_simuling/LLC/Robot_Plant/EE/H2/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/H3/Solid
- DIY_simuling/LLC/Robot_Plant/EE/H3/Transform
- DIY_simuling/LLC/Robot_Plant/EE/H3/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/Hand/Solid
- DIY_simuling/LLC/Robot_Plant/EE/Hand/Transform
- DIY_simuling/LLC/Robot_Plant/EE/Hand/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/Hand/Transform2
- DIY_simuling/LLC/Robot_Plant/EE/Hand/Transform3
- DIY_simuling/LLC/Robot_Plant/EE/Revolute1
- DIY_simuling/LLC/Robot_Plant/EE/Revolute10
- DIY_simuling/LLC/Robot_Plant/EE/Revolute11
- DIY_simuling/LLC/Robot_Plant/EE/Revolute6

- DIY_simuling/LLC/Robot_Plant/EE/Revolute7
- DIY_simuling/LLC/Robot_Plant/EE/Revolute8
- DIY_simuling/LLC/Robot_Plant/EE/Revolute9
- DIY_simuling/LLC/Robot_Plant/EE/cap/Solid
- DIY_simuling/LLC/Robot_Plant/EE/cap/Transform
- DIY_simuling/LLC/Robot_Plant/EE/cap/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/cap/Transform2
- DIY_simuling/LLC/Robot_Plant/EE/cap/Transform3
- DIY_simuling/LLC/Robot_Plant/EE/main_hand/Solid
- DIY_simuling/LLC/Robot_Plant/EE/main_hand/Transform
- DIY_simuling/LLC/Robot_Plant/EE/main_hand/Transform1
- DIY_simuling/LLC/Robot_Plant/EE/main_hand/Transform2
- DIY_simuling/LLC/Robot_Plant/EE/main_hand/Transform3
- DIY_simuling/LLC/Robot_Plant/EE/scrow/Solid
- DIY_simuling/LLC/Robot_Plant/EE/scrow/Transform
- DIY_simuling/LLC/Robot_Plant/J1
- DIY_simuling/LLC/Robot_Plant/J2
- DIY_simuling/LLC/Robot_Plant/J3
- DIY_simuling/LLC/Robot_Plant/J4
- DIY_simuling/LLC/Robot_Plant/J5
- DIY_simuling/LLC/Robot_Plant/Link1/Solid
- DIY_simuling/LLC/Robot_Plant/Link1/Transform
- DIY_simuling/LLC/Robot_Plant/Link1/Transform1
- DIY_simuling/LLC/Robot_Plant/Link1_2/Solid
- DIY_simuling/LLC/Robot_Plant/Link1_2/Transform
- DIY_simuling/LLC/Robot_Plant/Link1_2/Transform1
- DIY_simuling/LLC/Robot_Plant/Link1_2/Transform2
- DIY_simuling/LLC/Robot_Plant/Link1_4/Solid
- DIY_simuling/LLC/Robot_Plant/Link1_4/Transform
- DIY_simuling/LLC/Robot_Plant/Link1_4/Transform1
- DIY_simuling/LLC/Robot_Plant/Link2_2/Solid
- DIY_simuling/LLC/Robot_Plant/Link2_2/Transform
- DIY_simuling/LLC/Robot_Plant/Link3/Solid
- DIY_simuling/LLC/Robot_Plant/Link3/Transform
- DIY_simuling/LLC/Robot_Plant/Link3/Transform1
- DIY_simuling/LLC/Robot_Plant/Transform
- DIY_simuling/LLC/Robot_Plant/link2_4/Solid
- DIY_simuling/LLC/Robot_Plant/link2_4/Transform
- DIY_simuling/LLC/Robot_Plant/link2_4/Transform1

Resolved in: model workspace (DIY_simuling)

Chapter 5. Requirements

DIY_simuling does not contain requirements traceability links.

Chapter 6. System Model Configuration

Source: Model
Source Name: DIY_simuling

Table 6.1. DIY_simuling Configuration Set

Property	Value
Description	
Components	[DIY_simuling Configuration Set.Components(1), DIY_simuling Configuration Set.Components(2), DIY_simuling Configuration Set.Components(3), DIY_simuling Configuration Set.Components(4), DIY_simuling Configuration Set.Components(5), DIY_simuling Configuration Set.Components(6), DIY_simuling Configuration Set.Components(7), DIY_simuling Configuration Set.Components(8), DIY_simuling Configuration Set.Components(9), DIY_simuling Configuration Set.Components(10), DIY_simuling Configuration Set.Components(11)]
Name	Configuration
SimulationMode	normal
ConfigType	Model

Table 6.2. DIY_simuling Configuration Set.Components(1)

Property	Value
Name	Solver
Description	
Components	
StartTime	0.0
StopTime	10
AbsTol	auto
AutoScaleAbsTol	on
FixedStep	auto
InitialStep	auto
MaxNumMinSteps	-1
MaxOrder	5
ZcThreshold	auto
ConsecutiveZCsStepRelTol	10*128*eps

MaxConsecutiveZCs	1000
ExtrapolationOrder	4
NumberNewtonIterations	1
MaxStep	0.2
MinStep	auto
MaxConsecutiveMinStep	1
RelTol	1e-3
SolverMode	SingleTasking
EnableMultiTasking	off
EnableExplicitPartitioning	off
EnableConcurrentExecution	on
ConcurrentTasks	off
Solver	ode23t
SolverName	ode23t
SolverType	Variable-step
SolverJacobianMethodControl	auto
ShapePreserveControl	DisableAll
ZeroCrossControl	UseLocalSettings
ZeroCrossAlgorithm	Nonadaptive
SolverResetMethod	Fast
PositivePriorityOrder	off
AutoInsertRateTranBlk	off
SampleTimeConstraint	Unconstrained
InsertRTBMode	Whenever possible
SampleTimeProperty	
DecoupledContinuousIntegration	off
MinimalZcImpactIntegration	off
SolverOrder	3

Table 6.3. DIY_simuling Configuration Set.Components(2)

Property	**Value**
Name	Data Import/Export
Description	
Components	
Decimation	1
ExternalInput	[t, u]
FinalStateName	xFinal

InitialState	xInitial
LimitDataPoints	off
MaxDataPoints	1000
LoadExternalInput	off
LoadInitialState	off
SaveFinalState	off
SaveCompleteFinalSimState	off
SaveOperatingPoint	off
SaveFormat	Dataset
SaveOutput	on
SaveState	off
SignalLogging	on
DSMLogging	on
InspectSignalLogs	off
SaveTime	on
ReturnWorkspaceOutputs	on
StateSaveName	xout
TimeSaveName	tout
OutputSaveName	yout
SignalLoggingName	logsout
DSMLoggingName	dsmout
OutputOption	RefineOutputTimes
OutputTimes	[]
ReturnWorkspaceOutputsName	out
Refine	1
LoggingToFile	off
DatasetSignalFormat	timeseries
LoggingFileName	out.mat
LoggingIntervals	[-inf, inf]

Table 6.4. DIY_simuling Configuration Set.Components(3)

Property	**Value**
Name	Optimization
Description	
Components	
BlockReduction	on
BooleanDataType	on

ConditionallyExecuteInputs	on
DefaultParameterBehavior	Tunable
InlineParams	off
UseDivisionForNetSlopeComputation	off
GainParamInheritBuiltInType	off
UseFloatMulNetSlope	off
DefaultUnderspecifiedDataType	double
UseSpecifiedMinMax	off
InlineInvariantSignals	off
OptimizeBlockIOStorage	on
BufferReuse	on
GlobalBufferReuse	on
GlobalVariableUsage	None
StrengthReduction	off
AdvancedOptControl	
EnforceIntegerDowncast	on
ExpressionFolding	on
BooleansAsBitfields	off
BitfieldContainerType	uint_T
EnableMemcpy	on
MemcpyThreshold	64
PassReuseOutputArgsAs	Structure reference
PassReuseOutputArgsThreshold	12
FoldNonRolledExpr	on
LocalBlockOutputs	on
RollThreshold	5
StateBitsets	off
DataBitsets	off
ActiveStateOutputEnumStorageType	Native Integer
UseTempVars	off
ZeroExternalMemoryAtStartup	on
ZeroInternalMemoryAtStartup	on
InitFltsAndDblsToZero	off
NoFixptDivByZeroProtection	off
EfficientFloat2IntCast	off
EfficientMapNaN2IntZero	on
LifeSpan	auto
EvaledLifeSpan	Inf

MaxStackSize	Inherit from target
BufferReusableBoundary	on
SimCompilerOptimization	off
AccelVerboseBuild	off
OptimizeBlockOrder	off
OptimizeDataStoreBuffers	on
BusAssignmentInplaceUpdate	on
DifferentSizesBufferReuse	off
OptimizationLevel	level2
OptimizationPriority	Balanced
OptimizationCustomize	on
UseRowMajorAlgorithm	off
LabelGuidedReuse	off
MultiThreadedLoops	off
DenormalBehavior	GradualUnderflow

Table 6.5. DIY_simuling Configuration Set.Components(4)

Property	Value
Name	Diagnostics
Description	
Components	
RTPrefix	error
ConsistencyChecking	none
ArrayBoundsChecking	none
SignalInfNanChecking	none
StringTruncationChecking	error
SignalRangeChecking	none
ReadBeforeWriteMsg	UseLocalSettings
WriteAfterWriteMsg	UseLocalSettings
WriteAfterReadMsg	UseLocalSettings
AlgebraicLoopMsg	warning
ArtificialAlgebraicLoopMsg	warning
SaveWithDisabledLinksMsg	warning
SaveWithParameterizedLinksMsg	warning
CheckSSInitialOutputMsg	on
UnderspecifiedInitializationDetection	Simplified
MergeDetectMultiDrivingBlocksExec	error

CheckExecutionContextRuntimeOutputMsg	off
SignalResolutionControl	UseLocalSettings
BlockPriorityViolationMsg	warning
MinStepSizeMsg	warning
TimeAdjustmentMsg	none
MaxConsecutiveZCsMsg	error
MaskedZcDiagnostic	warning
IgnoredZcDiagnostic	warning
SolverPrmCheckMsg	none
InheritedTsInSrcMsg	warning
MultiTaskDSMMsg	error
MultiTaskCondExecSysMsg	error
MultiTaskRateTransMsg	error
SingleTaskRateTransMsg	none
TasksWithSamePriorityMsg	warning
SigSpecEnsureSampleTimeMsg	warning
CheckMatrixSingularityMsg	none
IntegerOverflowMsg	warning
Int32ToFloatConvMsg	warning
ParameterDowncastMsg	error
ParameterOverflowMsg	error
ParameterUnderflowMsg	none
ParameterPrecisionLossMsg	warning
ParameterTunabilityLossMsg	warning
FixptConstUnderflowMsg	none
FixptConstOverflowMsg	none
FixptConstPrecisionLossMsg	none
UnderSpecifiedDataTypeMsg	none
UnnecessaryDatatypeConvMsg	none
VectorMatrixConversionMsg	none
FcnCallInpInsideContextMsg	error
SignalLabelMismatchMsg	none
UnconnectedInputMsg	warning
UnconnectedOutputMsg	warning
UnconnectedLineMsg	warning
UseOnlyExistingSharedCode	error
SFcnCompatibilityMsg	none
FrameProcessingCompatibilityMsg	error

UniqueDataStoreMsg	none
BusObjectLabelMismatch	warning
RootOutportRequireBusObject	warning
AssertControl	UseLocalSettings
Echo	
EnableOverflowDetection	off
AllowSymbolicDim	on
ModelReferenceIOMsg	none
ModelReferenceVersionMismatchMessage	none
ModelReferenceIOMismatchMessage	none
ModelReferenceCSMismatchMessage	none
ModelReferenceSimTargetVerbose	off
UnknownTsInhSupMsg	warning
ModelReferenceDataLoggingMessage	warning
ModelReferenceSymbolNameMessage	warning
ModelReferenceExtraNoncontSigs	error
StateNameClashWarn	none
OperatingPointInterfaceChecksumMismatchMsg	warning
NonCurrentReleaseOperatingPointMsg	error
PregeneratedLibrarySubsystemCodeDiagnostic	warning
InitInArrayFormatMsg	warning
StrictBusMsg	ErrorLevel1
BusNameAdapt	WarnAndRepair
NonBusSignalsTreatedAsBus	none
SFUnusedDataAndEventsDiag	warning
SFUnexpectedBacktrackingDiag	error
SFInvalidInputDataAccessInChartInitDiag	warning
SFNoUnconditionalDefaultTransitionDiag	error
SFTransitionOutsideNaturalParentDiag	warning
SFUnconditionalTransitionShadowingDiag	warning
SFUnreachableExecutionPathDiag	warning
SFUndirectedBroadcastEventsDiag	warning
SFTransitionActionBeforeConditionDiag	warning
SFOutputUsedAsStateInMooreChartDiag	error
SFTemporalDelaySmallerThanSampleTimeDiag	warning
SFUnconditionalPathOutOfParentDiag	warning
SFSelfTransitionDiag	warning
SFExecutionAtInitializationDiag	warning

SFMachineParentedDataDiag	warning
SFUnreachableStateOrJunctionDiag	warning
SFDanglingTransitionDiag	warning
IntegerSaturationMsg	warning
AllowedUnitSystems	all
UnitsInconsistencyMsg	warning
AllowAutomaticUnitConversions	on
RCSCRenamedMsg	warning
RCSCObservableMsg	warning
ForceCombineOutputUpdateInSim	off
UnderSpecifiedDimensionMsg	none
DebugExecutionForFMUViaOutOfProcess	off
ArithmeticOperatorsInVariantConditions	error

Table 6.6. DIY_simuling Configuration Set.Components(5)

Property	Value
Name	Hardware Implementation
Description	
Components	
ProdBitPerChar	8
ProdBitPerShort	16
ProdBitPerInt	32
ProdBitPerLong	32
ProdBitPerLongLong	64
ProdBitPerFloat	32
ProdBitPerDouble	64
ProdBitPerPointer	64
ProdBitPerSizeT	64
ProdBitPerPtrDiffT	64
ProdLargestAtomicInteger	Char
ProdLargestAtomicFloat	Float
ProdIntDivRoundTo	Zero
ProdEndianess	LittleEndian
ProdWordSize	64
ProdShiftRightIntArith	on
ProdLongLongMode	off
ProdHWDeviceType	Intel->x86-64 (Windows64)

TargetBitPerChar	8
TargetBitPerShort	16
TargetBitPerInt	32
TargetBitPerLong	32
TargetBitPerLongLong	64
TargetBitPerFloat	32
TargetBitPerDouble	64
TargetBitPerPointer	32
TargetBitPerSizeT	32
TargetBitPerPtrDiffT	32
TargetLargestAtomicInteger	Char
TargetLargestAtomicFloat	None
TargetShiftRightIntArith	on
TargetLongLongMode	off
TargetIntDivRoundTo	Undefined
TargetEndianess	Unspecified
TargetWordSize	32
TargetPreprocMaxBitsSint	32
TargetPreprocMaxBitsUint	32
TargetHWDeviceType	Specified
TargetUnknown	off
ProdEqTarget	on
UseEmbeddedCoderFeatures	on
UseSimulinkCoderFeatures	on
HardwareBoardFeatureSet	EmbeddedCoderHSP

Table 6.7. DIY_simuling Configuration Set.Components(6)

Property	**Value**
Name	Model Referencing
Description	
Components	
UpdateModelReferenceTargets	IfOutOfDateOrStructuralChange
SkipRefExpFcnMdlSchedulingOrderCheck	off
EnableRefExpFcnMdlSchedulingChecks	on
CheckModelReferenceTargetMessage	error
EnableParallelModelReferenceBuilds	off
ParallelModelReferenceErrorOnInvalidPool	on

ParallelModelReferenceMATLABWorkerInit	None
ModelReferenceNumInstancesAllowed	Multi
PropagateVarSize	Infer from blocks in model
ModelDependencies	
ModelReferencePassRootInputsByReference	on
ModelReferenceMinAlgLoopOccurrences	off
PropagateSignalLabelsOutOfModel	on
SupportModelReferenceSimTargetCustomCode	off

Table 6.8. DIY_simuling Configuration Set.Components(7)

Property	**Value**
Name	Simulation Target
Description	
Components	
SimCustomSourceCode	
SimCustomHeaderCode	
SimCustomInitializer	
SimCustomTerminator	
SimReservedNameArray	
SimUserSources	
SimUserIncludeDirs	
SimUserLibraries	
SimUserDefines	
SFSimEnableDebug	off
SFSimOverflowDetection	on
SFSimEcho	on
SimBlas	on
SimCtrlC	on
SimExtrinsic	on
SimIntegrity	on
SimUseLocalCustomCode	on
SimParseCustomCode	on
SimAnalyzeCustomCode	off
SimBuildMode	sf_incremental_build
SimDataInitializer	
SimGenImportedTypeDefs	off
CompileTimeRecursionLimit	50

EnableRuntimeRecursion	on
MATLABDynamicMemAlloc	on
MATLABDynamicMemAllocThreshold	65536
CustomSymbolStrEMXArray	nothing
CustomSymbolStrEMXArrayFcn	nothing
CustomCodeFunctionArrayLayout	
DefaultCustomCodeFunctionArrayLayout	NotSpecified

Table 6.9. DIY_simuling Configuration Set.Components(8)

Property	Value
Name	Code Generation
SystemTargetFile	grt.tlc
HardwareBoard	None
ShowCustomHardwareApp	off
ShowEmbeddedHardwareApp	off
TLCOptions	
CodeGenDirectory	
GenCodeOnly	off
MakeCommand	make_rtw
GenerateMakefile	on
PackageGeneratedCodeAndArtifacts	off
PackageName	
TemplateMakefile	grt_default_tmf
PostCodeGenCommand	
Description	
GenerateReport	off
SaveLog	off
RTWVerbose	on
RetainRTWFile	off
ProfileTLC	off
TLCDebug	off
TLCCoverage	off
TLCAssert	off
ProcessScriptMode	Default
ConfigurationMode	Optimized
ProcessScript	
ConfigurationScript	

ConfigAtBuild	off
RTWUseLocalCustomCode	on
RTWUseSimCustomCode	off
CustomSourceCode	
CustomHeaderCode	
CustomInclude	
CustomSource	
CustomLibrary	
CustomDefine	
CustomBLASCallback	
CustomLAPACKCallback	
CustomFFTCallback	
CustomInitializer	
CustomTerminator	
Toolchain	Automatically locate an installed toolchain
BuildConfiguration	Faster Builds
CustomToolchainOptions	
IncludeHyperlinkInReport	off
LaunchReport	off
RecursionLimit	50
PortableWordSizes	off
GenerateErtSFunction	off
CreateSILPILBlock	None
CodeExecutionProfiling	off
CodeExecutionProfileVariable	executionProfile
CodeProfilingSaveOptions	SummaryOnly
CodeProfilingInstrumentation	off
CodeCoverageSettings	DIY_simuling Configuration Set.Components(8).CodeCoverageSettings
SILDebugging	off
TargetLang	C
IncludeERTFirstTime	off
GenerateTraceInfo	off
GenerateTraceReport	off
GenerateTraceReportSl	off
GenerateTraceReportSf	off
GenerateTraceReportEml	off
GenerateCodeInfo	off

GenerateWebview	off
GenerateCodeMetricsReport	off
GenerateCodeReplacementReport	off
RTWCompilerOptimization	off
ObjectivePriorities	
RTWCustomCompilerOptimizations	
CheckMdlBeforeBuild	Off
CustomRebuildMode	OnUpdate
DataInitializer	
Components	[DIY_simuling Configuration Set.Components(8).Components(1), DIY_simuling Configuration Set.Components(8).Components(2)]

Table 6.10. DIY_simuling Configuration Set.Components(9)

Property	**Value**
Description	Simulink Coverage Configuration Component
Components	
Name	Simulink Coverage
CovEnable	off
CovScope	EntireSystem
CovIncludeTopModel	on
RecordCoverage	off
CovPath	/
CovSaveName	covdata
CovCompData	
CovMetricSettings	dwe
CovFilter	
CovHTMLOptions	
CovNameIncrementing	off
CovHtmlReporting	off
CovForceBlockReductionOff	on
CovEnableCumulative	on
CovSaveCumulativeToWorkspaceVar	off
CovSaveSingleToWorkspaceVar	off
CovCumulativeVarName	covCumulativeData
CovCumulativeReport	off
CovSaveOutputData	on

CovOutputDir	slcov_output/$ModelName$
CovDataFileName	$ModelName$_cvdata
CovShowResultsExplorer	on
CovReportOnPause	on
CovModelRefEnable	off
CovModelRefExcluded	
CovExternalEMLEnable	on
CovSFcnEnable	on
CovBoundaryAbsTol	1.0000e-05
CovBoundaryRelTol	0.0100
CovUseTimeInterval	off
CovStartTime	0
CovStopTime	0
CovMetricStructuralLevel	Decision
CovMetricLookupTable	off
CovMetricSignalRange	off
CovMetricSignalSize	off
CovMetricObjectiveConstraint	off
CovMetricSaturateOnIntegerOverflow	off
CovMetricRelationalBoundary	off
CovLogicBlockShortCircuit	off
CovUnsupportedBlockWarning	on
CovHighlightResults	off
CovMcdcMode	Masking

Table 6.11. DIY_simuling Configuration Set.Components(10)

Property	Value
Description	HDL Coder custom configuration component
Components	
Name	HDL Coder

Table 6.12. DIY_simuling Configuration Set.Components(11)

Property	Value
Description	
Components	[DIY_simuling Configuration Set.Components(11).Components(1), DIY_simuling Configuration Set.Components(11).Components(2)]

Name	Simscape
EditingMode	Full
ExplicitSolverDiagnosticOptions	warning
GlobalZcOffDiagnosticOptions	warning
SimscapeNormalizeSystem	on
SimscapeNominalValues	[{"value":"1","unit":"A"},{"value":"1","unit":"bar"},{"value":"1","unit":"cm^2"},{"value":"1","unit":"cm^3/s"},{"value":"1","unit":"kJ/kg"},{"value":"1","unit":"kW"},{"value":"1","unit":"l"},{"value":"1","unit":"N"},{"value":"1","unit":"N*m"},{"value":"1","unit":"V"}]
SimscapeLogType	none
SimscapeLogSimulationStatistics	off
SimscapeLogToSDI	off
SimscapeLogOpenViewer	off
SimscapeLogName	simlog
SimscapeLogDecimation	1
SimscapeLogLimitData	on
SimscapeLogDataHistory	5000
SimscapeUseOperatingPoints	off
SimscapeOperatingPoint	
SelectedTab	
Version	1.0
ComponentsAttached	true
Listener	[DIY_simuling Configuration Set.Components(11).Listener(1), DIY_simuling Configuration Set.Components(11).Listener(2), DIY_simuling Configuration Set.Components(11).Listener(3), DIY_simuling Configuration Set.Components(11).Listener(4), DIY_simuling Configuration Set.Components(11).Listener(5), DIY_simuling Configuration Set.Components(11).Listener(6), DIY_simuling Configuration Set.Components(11).Listener(7), DIY_simuling Configuration Set.Components(11).Listener(8), DIY_simuling Configuration Set.Components(11).Listener(9), DIY_simuling Configuration Set.Components(11).Listener(10), DIY_simuling Configuration Set.Components(11).Listener(11), DIY_simuling Configuration Set.Components(11).Listener(12), DIY_simuling Configuration Set.Components(11).Listener(13), DIY_simuling Configuration Set.Components(11).Listener(14), DIY_simuling Configuration Set.Components(11).Listener(15), DIY_simuling Configuration Set.Components(11).Listener(16)]
someListenersNotInstalled	false
instanceId	

Table 6.13. DIY_simuling Configuration Set.Components(8).CodeCoverageSettings

Property	Value
TopModelCoverage	off
ReferencedModelCoverage	off
CoverageTool	None

Table 6.14. DIY_simuling Configuration Set.Components(8).Components(1)

Property	Value
Name	Code Appearance
Description	
Components	
ForceParamTrailComments	off
GenerateComments	on
CommentStyle	Auto
IgnoreCustomStorageClasses	on
IgnoreTestpoints	off
IncHierarchyInIds	off
MaxIdLength	31
ShowEliminatedStatement	off
OperatorAnnotations	off
IncAutoGenComments	off
SimulinkDataObjDesc	off
SFDataObjDesc	off
MATLABFcnDesc	off
IncDataTypeInIds	off
PrefixModelToSubsysFcnNames	on
MangleLength	1
SharedChecksumLength	8
CustomSymbolStr	RN$M
CustomSymbolStrGlobalVar	RN$M
CustomSymbolStrType	NR$M_T
CustomSymbolStrField	NM
CustomSymbolStrFcn	RNMF
CustomSymbolStrSimulinkFcn	RN
CustomSymbolStrFcnArg	rt$I$N$M

CustomSymbolStrBlkIO	rtb_NM
CustomSymbolStrTmpVar	NM
CustomSymbolStrMacro	RN$M
CustomSymbolStrUtil	NC
CustomSymbolStrEmxType	emxArray_MN
CustomSymbolStrEmxFcn	emx$M$N
CustomUserTokenString	
CustomCommentsFcn	
DefineNamingRule	None
DefineNamingFcn	
ParamNamingRule	None
ParamNamingFcn	
SignalNamingRule	None
SignalNamingFcn	
InsertBlockDesc	off
InsertPolySpaceComments	off
SimulinkBlockComments	on
BlockCommentType	BlockPathComment
StateflowObjectComments	off
MATLABSourceComments	off
EnableCustomComments	off
InternalIdentifier	Shortened
InlinedPrmAccess	Literals
ReqsInCode	off
UseSimReservedNames	off
ReservedNameArray	

Table 6.15. DIY_simuling Configuration Set.Components(8).Components(2)

Property	**Value**
Name	Target
Description	
Components	
IsERTTarget	off
TargetLibSuffix	
TargetPreCompLibLocation	
GenFloatMathFcnCalls	NOT IN USE
TargetLangStandard	C99 (ISO)

TargetFunctionLibrary	NOT IN USE
CodeReplacementLibrary	None
UtilityFuncGeneration	Auto
MultiwordTypeDef	System defined
MultiwordLength	2048
DynamicStringBufferSize	256
GenerateFullHeader	on
InferredTypesCompatibility	off
ExistingSharedCode	
SharedCodeLocation	
GenerateSampleERTMain	off
GenerateTestInterfaces	off
ModelReferenceCompliant	on
ParMdlRefBuildCompliant	on
CompOptLevelCompliant	on
ConcurrentExecutionCompliant	on
IncludeMdlTerminateFcn	on
CombineOutputUpdateFcns	on
CombineSignalStateStructs	off
GroupInternalDataByFunction	off
SuppressErrorStatus	off
ERTFirstTimeCompliant	off
IncludeFileDelimiter	Auto
ERTCustomFileBanners	off
SupportAbsoluteTime	on
LogVarNameModifier	rt_
MatFileLogging	on
MultiInstanceERTCode	off
CodeInterfacePackaging	Nonreusable function
PurelyIntegerCode	off
SupportNonFinite	on
SupportComplex	on
SupportContinuousTime	on
SupportNonInlinedSFcns	on
RemoveDisableFunc	off
RemoveResetFunc	off
SupportVariableSizeSignals	off
ParenthesesLevel	Nominal

CastingMode	Nominal
PreserveStateflowLocalDataDimensions	off
GenerateClassInterface	off
ModelStepFunctionPrototypeControlCompliant	off
CPPClassGenCompliant	on
GRTInterface	off
GenerateAllocFcn	off
UseToolchainInfoCompliant	on
GenerateSharedConstants	on
LUTObjectStructOrderExplicitValues	Size,Breakpoints,Table
LUTObjectStructOrderEvenSpacing	Size,Breakpoints,Table
ArrayLayout	Column-major
UnsupportedSFcnMsg	error
ERTHeaderFileRootName	RE
ERTSourceFileRootName	RE
ERTDataFileRootName	$R_data
ExtMode	off
ExtModeStaticAlloc	off
ExtModeTesting	off
ExtModeStaticAllocSize	1000000
ExtModeTransport	0
ExtModeMexFile	ext_comm
ExtModeMexArgs	
ExtModeIntrfLevel	Level1
RTWCAPISignals	off
RTWCAPIParams	off
RTWCAPIStates	off
RTWCAPIRootIO	off
GenerateASAP2	off
MultiInstanceErrorCode	Error

Table 6.16. DIY_simuling Configuration Set.Components(11).Components(1)

Property	Value
Description	
Components	
Name	SimscapeMultibody1G
WarnOnRedundantConstraints	on

WarnOnSingularInitialAssembly	off
ShowCutJoints	off
VisOnUpdateDiagram	off
VisDuringSimulation	off
EnableVisSimulationTime	on
VisSampleTime	0
DisableBodyVisControl	off
ShowCG	on
ShowCS	on
ShowOnlyPortCS	off
HighlightModel	on
FramesToBeSkipped	0
AnimationDelay	3
RecordAVI	off
CompressAVI	on
AviFileName	
AutoFitVis	off
EnableSelection	on
LastVizWinPosition	[-1 -1 -1 -1]
CamPosition	[0 0 0]
CamTarget	[0 0 -1]
CamUpVector	[0 1 0]
CamHeight	-1
CamViewAngle	0
VisBackgroundColor	[0.9 0.9 0.95]
DefaultBodyColor	[1 0 0]
MDLBodyVisualizationType	Convex hull from body CS locations
OVRRIDBodyVisualizationType	NONE
VisConfigFile	

Table 6.17. DIY_simuling Configuration Set.Components(11).Components(2)

Property	Value
Description	Simscape Multibody
Components	[DIY_simuling Configuration Set.Components(11).Components(2).Components(1), DIY_simuling Configuration Set.Components(11).Components(2).Components(2)]
Name	SimscapeMultibody

DIY_simuling Configuration Set.Components(11).Listener(1) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(2) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(3) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(4) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(5) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(6) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(7) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(8) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(9) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(10) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(11) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(12) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(13) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(14) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(15) (handle.listener,)

Note: this object has no unfiltered properties.

DIY_simuling Configuration Set.Components(11).Listener(16) (handle.listener,)

Note: this object has no unfiltered properties.

Table 6.18. DIY_simuling Configuration Set.Components(11).Components(2).Components(1)

Property	Value
Description	Diagnostics
Components	
Name	DiagnosticsConfigSet
SimMechanicsInvalidVisualProperty	warning
SimMechanicsCrossSectionNullEdge	warning
SimMechanicsUnconnectedFramePorts	warning
SimMechanicsUnconnectedGeometryPorts	warning
SimMechanicsRedundantBlock	warning
SimMechanicsConflictingReferenceFrames	warning
SimMechanicsRigidlyBoundBlock	error
SimMechanicsUnsatisfiedHighPriorityTargets	warning
SimMechanicsJointTargetOverSpecification	error

Table 6.19. DIY_simuling Configuration Set.Components(11).Components(2).Components(2)

Property	Value
Description	Explorer
Components	
Name	ExplorerConfigSet
SimMechanicsOpenEditorOnUpdate	on
InternalSimMechanicsExplorerSettings	

Table 6.20. HDL Coder

Property	Value
HDLSubsystem	DIY_simuling
Workflow	Generic ASIC/FPGA
TargetPlatform	

ReferenceDesign	
ReferenceDesignPath	
CoeffPrefix	coeff
InputType	std_logic_vector
OutputType	Same as input type
ScalarizePorts	off
CoeffMultipliers	Multiplier
ResetType	Asynchronous
FIRAdderStyle	linear
MultiplierInputPipeline	0
MultiplierOutputPipeline	0
FoldingFactor	1
NumMultipliers	-1
OptimizeForHDL	off
TimingControllerPostfix	_tc
OptimizeTimingController	on
TimingControllerArch	default
CastBeforeSum	on
TCCounterLimitCompOp	>=
CheckHDL	off
EnablePrefix	enb
ClockEnableInputPort	clk_enable
ClockEnableOutputPort	ce_out
ClockInputPort	clk
ClockEdge	Rising
ResetInputPort	reset
SimulatorFlags	
HDLCompileFilePostfix	_compile.do
HDLCompileInit	vlib %s\n
HDLCompileTerm	
HDLCompileVerilogCmd	vlog %s %s\n
HDLCompileVHDLCmd	vcom %s %s\n
EnableForGenerateLoops	on
HDLMapFilePostfix	_map.txt
HDLMapSeparator	
HDLSimCmd	vsim -novopt %s.%s\n
HDLSimFilePostfix	_sim.do
HDLSimProjectFilePostfix	_init.do

HDLSimInit	onbreak resume\nonerror resume\n
HDLSimProjectCmd	project addfile %s\n
HDLSimProjectTerm	project compileall\n
HDLSimProjectInit	project new . %s work\n
HDLSimTerm	run -all\n
HDLSimViewWaveCmd	add wave sim:%s\n
HDLSynthTool	None
HDLSynthCmd	
HDLSynthFilePostfix	
HDLSynthInit	
HDLSynthLibCmd	
HDLSynthLibSpec	
HDLSynthTerm	
ReservedWordPostfix	_rsvd
BlockGenerateLabel	_gen
VHDLLibraryName	work
UseSingleLibrary	off
VHDLArchitectureName	rtl
ClockProcessPostfix	_process
ComplexImagPostfix	_im
ComplexRealPostfix	_re
EntityConflictPostfix	_block
InstancePrefix	u_
InstancePostfix	
InstanceGenerateLabel	_gen
OutputGenerateLabel	outputgen
PackagePostfix	_pkg
SplitEntityArch	off
SplitEntityFilePostfix	_entity
SplitArchFilePostfix	_arch
VectorPrefix	vector_of_
ClockInputs	Single
TriggerAsClock	off
ConditionalizePipeline	off
InferControlPorts	off
UseRisingEdge	off
TargetDirectory	hdlsrc
TargetSubdirectory	Model

EDAScriptGeneration	on
AddInputRegister	on
AddOutputRegister	on
AddPipelineRegisters	off
PipelinePostfix	_pipe
InputPort	filter_in
OutputPort	filter_out
FracDelayPort	filter_fd
Name	filter
RemoveResetFrom	None
ResetAssertedLevel	Active-high
ReuseAccum	off
ScaleWarnBits	3
SerialPartition	-1
DALUTPartition	-1
DARadix	2
CoefficientSource	Internal
CoefficientMemory	Registers
InputComplex	off
AddRatePort	off
InputDataType	
GenerateHDLCode	on
GenerateModel	on
GenerateTB	off
GenerateCEGenModel	off
ObfuscateGeneratedHDLCode	off
Traceability	off
ResourceReport	off
OptimizationReport	off
ErrorCheckReport	on
HDLGenerateWebview	off
IPCoreReport	off
Recommendations	off
RequirementComments	on
Backannotation	off
HierarchicalDistPipelining	off
PreserveDesignDelays	off
AcquireDesignDelaysForEMLOptimizations	off

ClockRatePipelining	on
CRPWithoutFlattening	on
UseCRPAlternativeStrategy	off
IncreaseCRPBudget	on
AdaptivePipelining	on
MinDelaysRequiredAtLocalMultirateOutput	1
ClockRatePipelineOutputPorts	off
CriticalPathEstimation	off
StaticLatencyPathAnalysis	off
optimizeserializer	on
shareequalwl	on
sharedmulsign	Signed
MultiplierPromotionThreshold	0
RoutingFudgeFactor	0.5000
OptimizationCompatibilityCheck	off
NumCriticalPathsEstimated	1
CriticalPathEstimationFile	criticalPathEstimated
SLPAFile	staticLatPathAnalysis
SLPALoopsFile	staticLatLoops
HardwarePipeliningCharacterizationFile	
HighlightFeedbackLoops	on
HighlightFeedbackLoopsFile	highlightFeedbackLoop
HighlightClockRatePipeliningDiagnostic	on
HighlightClockRatePipeliningFile	highlightClockRatePipelining
DistributedPipeliningBarriers	on
DistributedPipeliningBarriersFile	highlightDistributedPipeliningBarriers
BlocksWithNoCharacterizationFile	highlightCriticalPathEstimationOffendingBlocks
AXIStreamingTransformFeatureControl	off
SerializerRatioThreshold	8192
RetimingCP	off
RetimingCPFile	highlightRetimingCP
ClearHighlightingFile	clearhighlighting
FunctionallyEquivalentRetiming	on
DistributedPipeliningPriority	Numerical Integrity
RetimingDetails	on
CriticalPathDetails	off
SignalNamesMangling	off
GuidedRetiming	off

LatencyConstraint	0
ReduceMatchingDelays	on
OptimizationData	
CPGuidanceFile	
CPAnnotationFile	
HandleAtomicSubsystem	on
OptimizeMdlGen	on
MulticyclePathInfo	off
MulticyclePathConstraints	off
MpswArchCaseWhen	off
FloatingPointTargetConfiguration	
GenerateTargetComps	on
NativeFloatingPoint	off
FPToleranceValue	1.0000e-07
FPToleranceStrategy	DEFAULT
nfpLatency	DEFAULT
nfpDenormals	DEFAULT
sschdlMatrixVectorProductEarlyElaborate	off
sschdlMatrixProductSumCustomLatency	-1
AlteraBackwardIncompatibleSinCosPipeline	off
FamilyDevicePackageSpeed	
ToolName	
SynthesisToolChipFamily	
SynthesisToolDeviceName	
SynthesisToolPackageName	
SynthesisToolSpeedValue	
SynthesisTool	
SynthesisProjectAdditionalFiles	
SimulationLibPath	
XilinxSimulatorLibPath	
AdderSharingMinimumBitwidth	0
MultiplierSharingMinimumBitwidth	0
MultiplyAddSharingMinimumBitwidth	0
ShareAdders	off
ShareMultipliers	on
ShareMultiplyAdds	on
ShareMATLABBlocks	on
ShareAtomicSubsystems	on

ShareFloatingPointIPs	on
PipelinedSharing	on
OptimizeCRPSharingRegisters	on
ClockRatePipeliningBudgetCheck	off
EnableFPGAWorkflow	off
FPGAWorkflowParameters	
GainMultipliers	Multiplier
ProductOfElementsStyle	linear
UserComment	
CustomFileHeaderComment	
CustomFileFooterComment	
DateComment	on
SafeZeroConcat	on
SumOfElementsStyle	linear
TargetLanguage	VHDL
Oversampling	1
ClockRatePipeliningFraction	1
Verbosity	1
TestBenchName	filter_tb
MultifileTestBench	off
IgnoreDataChecking	0
TestBenchPostfix	_tb
TestBenchDataPostfix	_data
TestBenchStimulus	
TestBenchUserStimulus	
TestBenchFracDelayStimulus	
TestBenchCoeffStimulus	
TestBenchRateStimulus	
ForceClockEnable	on
MinimizeClockEnables	off
MinimizeGlobalResets	off
NoResetInitializationMode	InsideModule
NoResetInitScript	noresetinitscript.tcl
ComplexMulElaboration	MultiplyAddBlock
FlattenBus	off
TestBenchClockEnableDelay	1
ForceClock	on
ClockHighTime	5

ClockLowTime	5
HoldTime	2
InputDataInterval	0
ForceReset	on
ErrorMargin	4
HoldInputDataBetweenSamples	on
InitializeTestBenchInputs	off
ResetLength	2
TestBenchReferencePostFix	_ref
GenerateValidationModel	off
RAMMappingThreshold	256
MapPipelineDelaysToRAM	off
RemoveRedundantCounters	on
ReplaceUnitDelayWithIntegerDelay	on
ConcatenateDelays	on
MergeDelaysOnFanouts	on
FoldDelaysToConstant	on
RAMArchitecture	WithClockEnable
InlineMATLABBlockCode	off
InlineHDLCode	off
MaskParameterAsGeneric	off
FlattenSharedSubsystems	off
StringTypeSupport	off
DeleteUnusedPorts	on
BalanceDelays	on
TargetFrequency	0
ExtraEffortMargin	1
MaxOversampling	Inf
MaxComputationLatency	1
MultiplierPartitioningThreshold	Inf
TreatDelayBalancingFailureAs	Error
TransformDelaysWithControlLogic	on
TransformNonZeroInitValDelay	on
DelayElaborationLimit	20
GenerateCoSimBlock	off
HDLCodeCoverage	off
GenerateHDLTestBench	on
GenerateCoSimModel	None

GenerateSVDPITestBench	None
SimulationTool	Mentor Graphics Modelsim
CoSimModelSetup	CosimBlockAndDut
SynthesisOnDirective	
SynthesisOffDirective	
LoopUnrolling	off
InlineConfigurations	on
UseAggregatesForConst	off
UseVerilogTimescale	on
Timescale	`timescale 1 ns / 1 ns
VerilogFileExtension	.v
SystemVerilogFileExtension	.sv
VHDLFileExtension	.vhd
CodeGenerationOutput	GenerateHDLCode
GeneratedModelName	
GeneratedModelNamePrefix	gm_
ValidationModelNameSuffix	_vnl
UseDotLayout	off
ShowCodeGenPIR	off
SerializeModel	0
SerializeIO	0
AutoRoute	on
AutoPlace	on
InterBlkHorzScale	1.7000
InterBlkVertScale	1.2000
CustomDotPath	
HighlightAncestors	on
HighlightColor	cyan
InitializeBlockRAM	on
InitializeRealPort	off
MapVectorPortToStream	off
UseFileIOInTestBench	on
TurnkeyWorkflow	off
AlteraWorkflow	off
GenerateFILBlock	off
CoSimLibPostfix	_cosim
TestBenchInitializeInputs	off
MinimizeIntermediateSignals	off

GenerateCodeInfo	off
GatewayoutWithDTC	off
IncrementalCodeGenForTopModel	off
HDLWFSmartbuild	on
HDLCodingStandard	None
HDLCodingStandardCustomizations	
ReferenceDesignParameter	
HDLLintTool	None
HDLLintInit	
HDLLintTerm	
HDLLintCmd	
ModulePrefix	
DetectBlackBoxNameCollision	Warning
PIRTB	on
PIRTC	off
EmitNetlist	off
UsePipelinedToolboxFunctions	on
savepirtoscript	off
ConcatenateHDLModules	off
AMS	off
ML2PIR	off
OptimBetweenMATLABAndSimulink	off
EnableTestpoints	off
TraceabilityStyle	Line Level
TreatRealsInGeneratedCodeAs	Error
EnumEncodingScheme	default
BuildToProtectModel	off
OptimizeConstants	on
StreamingMatrix	off
HDLDTO	off

Chapter 7. Glossary

Atomic Subsystem. A subsystem treated as a unit by an implementation of the design documented in this report. The implementation computes the outputs of all the blocks in the atomic subsystem before computing the next block in the parent system's block execution order (sorted list).

Block Diagram. A Simulink block diagram represents a set of simultaneous equations that relate a system or subsystem's inputs to its outputs as a function of time. Each block in the diagram represents an equation of the form y = f(t, x, u) where t is the current time, u is a block input, y is a block output, and x is a system state (see the Simulink documentation for information on the functions represented by the various types of blocks that make up the diagram). Lines connecting the blocks represent dependencies among the blocks, i.e., inputs whose current values are the outputs of other blocks. An implementation of a design described in this document computes a root or atomic system's outputs at each time step by computing the outputs of the blocks in an order determined by block input/output dependencies.

Block Parameter. A variable that determines the output of a block along with its inputs, for example, the gain parameter of a Gain block.

Block Execution Order. The order in which Simulink evaluates blocks during simulation of a model. The block execution order determined by Simulink ensures that a block executes only after all blocks on whose outputs it depends are executed.

Checksum. A number that indicates whether different versions of a model or atomic subsystem differ functionally or only cosmetically. Different checksums for different versions of the same model or subsystem indicate that the versions differ functionally.

Design Variable. A symbolic (MATLAB) variable or expression used as the value of a block parameter. Design variables allow the behavior of the model to be altered by altering the value of the design variable.

Signal. A block output, so-called because block outputs typically vary with time.

Virtual Subsystem. A subsystem that is purely graphical, i.e., is intended to reduce the visual complexity of the block diagram of which it is a subsystem. An implementation of the design treats the blocks in the subsystem as part of the first nonvirtual ancestor of the virtual subsystem (see Atomic Subsystem).

Chapter 8. About this Report

Report Overview

This report describes the design of the DIY_simuling system. The report was generated automatically from a Simulink model used to validate the design. It contains the following sections:

Model Version. Specifies information about the version of the model from which this design description was generated. Includes the model checksum, a number that indicates whether different versions of the model differ functionally or only cosmetically. Different checksums for different versions indicate that the versions differ functionally.

Root System. Describes the design's root system.

Subsystems. Describes each of the design's subsystems.

Design Variables. Describes system design variables, i.e., MATLAB variables and expressions used as block parameter values.

System Model Configuration. Lists the configuration parameters, e.g., start and stop time, of the model used to simulate the system described by this report.

Requirements. Shows design requirements associated with elements of the design model. This section appears only if the design model contains requirements links.

Glossary. Defines Simulink terms used in this report.

Root System Description

This section describes a design's root system. It contains the following sections:

Diagram. Simulink block diagram that represents the algorithm used to compute the root system's outputs.

Description. Description of the root system. This section appears only if the model's root system has a Documentation property or a Doc block.

Interface. Name, data type, width, and other properties of the root system's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the root system has input or output ports.

Blocks. This section has two subsections:

- **Parameters.** Describes key parameters of blocks in the root system. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, i.e., blocks that use lookup tables to compute their outputs.
- **Block Execution Order.** Order in which blocks must be executed at each time step in order to ensure that each block's inputs are available when it executes.

State Charts. Describes state charts used in the root system. This section appears only if the root system contains Stateflow blocks.

Subsystem Descriptions

This section describes a design's subsystems. Each subsystem description contains the following sections:

Checksum. This section appears only if the subsystem is an atomic subsystem. The checksum indicates whether the version of the model subsystem used to generate this report differs functionally from other versions of the model subsystem. If two model checksums differ, the corresponding versions of the model differ functionally.

Diagram. Simulink block diagram that graphically represents the algorithm used to compute the subsystem's outputs.

Description. Description of the subsystem. This section appears only if the subsystem has a Documentation property or contains a Doc block.

Interface. Name, data type, width, and other properties of the subsystem's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the subsystem is atomic and has input or output ports.

Blocks. Blocks that this subsystem contains. This section has two subsections:

- **Parameters.** Key parameters of blocks in the subsystem. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, blocks that use lookup tables to compute their outputs.
- **Block Execution Order.** Order in which the subsystem's blocks must be executed at each time step in order to ensure that each block's inputs are available when the block executes .This section appears only if the subsystem is atomic. Note: in Acrobat(PDF) reports, the number in square brackets next to the block name is a hyperlink to the block parameter table. The number has no model significance.

State Charts. Describes state charts used in the subsystem. This section appears only if the root system contains Stateflow blocks.

State Chart Descriptions

This section describes the state machines used by Stateflow blocks to compute their outputs, i.e., Stateflow blocks. Each state machine description contains the following sections:

Chart. Diagram representing the state machine.

States. Describes the state machine's states. Each state description includes the state's diagram and diagrams and/or descriptions of graphical functions, Simulink functions, truth tables, and MATLAB functions parented by the state.

Transitions. Transitions between the state machine's states. Each transition description specifies the values of key transition properties. Appears only if a transition has properties that do not appear on the chart.

Junctions. Transition junctions. Each junction description specifies the values of key junction properties. Appears only if a junction has properties that do not appear on the chart.

Events. Events that trigger state transitions. Each event description specifies the values of key event properties.

Data. Data types and other properties of the Stateflow block's inputs, outputs, and other state machine data.

Targets. Executable implementations of the state machine used to compute the outputs of the corresponding Stateflow block.

MATLAB Supporting Functions. List of functions invoked by MATLAB functions defined in the chart.